



**Sacramento River Bank Protection Project
Long-Term Aquatic Monitoring
Fish Sampling and Habitat Characterization
Fiscal Years 2011 through 2015**

Annual Report 2012 - Final

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Sacramento River Bank Protection Project Long-Term Aquatic Monitoring Program Fiscal Years 2011 through 2015

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Executive Summary

Major levee repairs and emergency levee repairs within the Sacramento River system have been traditionally conducted by the United States Army Corps of Engineers (USACE) through their Sacramento River Bank Protection Project (SRBPP), and the PL84-99 Rehabilitation Programs. ("PL 84-99" refers to federal Public Law 84-99, the Flood and Coastal Storm Emergencies Act). On February 24, 2006, Governor Arnold Schwarzenegger declared a State of Emergency for California's levee system, releasing up to \$500 million to repair and evaluate State and Federal project levees. These levee repair funds are now being administered through the existing USACE's levee repair programs and a new program, the Sacramento-San Joaquin Erosion Repairs Project (SSJERP). The USACE is the lead agency for the SRBPP and the PL 84-99 Rehabilitation Program, while the California State Department of Water Resources (DWR) is the lead agency for the SSJERP.

When the levees were originally constructed, maintained, and repaired (starting in the mid to late 19th century), mitigation measures were not required, but since the early 1990s, winter-run and spring-run Chinook salmon (*Oncorhynchus tshawytscha*), steelhead (*O. mykiss*), and the delta smelt (*Hypomesus transpacificus*) in the Sacramento River have been listed under the state and federal Endangered Species Acts. As a result, mitigation measures for channel and river bank modifications were developed for the post-2006 levee repairs and were designed based on the Standardized Assessment Methodology (SAM; USACE 2004, 2006; USACE and CVFPB 2009). The SAM is a predictive model used to quantify potential impacts to threatened and endangered fish species due to construction of bank revetment and other bank protection measures, and can also be utilized to assess the fish habitat value of experimental mitigation and compensation measures. The SAM quantifies habitat values in terms of a bank line or area-weighted species response index; habitat variables include bank slope, floodplain inundation ratio, bank substrate size, instream structure, aquatic vegetation, and overhanging shade.

Based on SAM results, four experimental levee repair designs were developed and constructed in the project area. For these post-2006 emergency levee repairs covered by the Governor's declaration, the National Marine Fisheries Service (NMFS) issued Biological Opinions (Opinion) pertaining to Chinook salmon and steelhead in June 2006 and August 2007; and the United States Fish and Wildlife Service (USFWS) issued Opinions pertaining to delta smelt in June 2006 and April 2007. The NMFS and USFWS Opinions support approval of critical levee repairs at 54 sites (along the Sacramento and Bear rivers; and Steamboat, Sutter, and Cache sloughs). DWR undertook repairs at 27 of the sites, and the USACE undertook repairs at the remaining 27 sites. These sites were identified as "critical," as defined by bank erosion that could threaten the structural integrity of the Sacramento River flood control system, and therefore required immediate work to prevent levee failure during the next flood (Ayres Associates 2007).

The NMFS and USFWS Opinions also required post-construction monitoring to evaluate whether "on-site compensations [at post-2006 emergency levee Repair Sites] are functioning in a manner that enhances habitat value and offset adverse bank protection effects" (NMFS 2001, 2004, 2006a, 2006b, 2007; USFWS 2004a, 2004b, 2006a, 2006b, 2006c, 2007). A fish and habitat monitoring plan (Sacramento River Bank Protection Project Long-Term Aquatic Monitoring Program) was developed in 2008 (H.T. Harvey & Associates et al. 2008) and revised

in 2011 (FISHBIO 2011), which established monitoring goals and objectives, as well as monitoring protocols to meet the post-construction monitoring requirements of these Opinions. Annual baseline monitoring has been conducted since 2008, though scope and methodology of monitoring have varied since initial collection of habitat and fisheries data began. Initial efforts to adequately characterize fish community composition and habitat use at selected repair sites suffered from low catch rates (largely attributable to low capture success using traps; H.T. Harvey & Associates and PRBO 2009), and the sampling methodology was subsequently revised to better document differences in abundance of focal fish species between different Mitigated Repair Types, naturalized sites and unmitigated repair sites. Single pass boat-based electrofishing was determined to be the most effective way to sample fish at the selected monitoring sites, and was implemented in 2009 (considered the first year in the baseline dataset for long-term monitoring evaluations). Since then the general approach to fish community sampling and subsequent data analysis has been relatively consistent, though minor adjustments have been made as pertaining to monitoring sites and field sampling methodology (outlined in more detail in subsequent sections).

This report documents and evaluates the results of the annual fish and habitat monitoring conducted in 2012, as well as comparisons to fish and habitat monitoring conducted from 2009-2010 (H.T. Harvey & Associates and PRBO 2011) and in 2011 (FISHBIO 2012). In 2012, a variety of habitat features were measured (described in more detail below), and fish presence and abundance was evaluated at several site types by electrofishing. Site types include post-2006 emergency levee repair sites that incorporate habitat mitigation features (Mitigated Repair Sites; $n = 11$); unmitigated pre-2006 levee repair sites consisting of rock revetment without additional habitat enhancement (Unmitigated Repair Sites; $n = 4$); and naturalized bank sites that have not been engineered, are not rip-rapped, and are dominated by native, naturally established vegetation (Naturalized Sites; $n = 7$). Through this monitoring effort, the following study objectives were addressed:

- 1) identify habitat features and design types of levee repair sites that promote use by focus species (anadromous salmonids and delta smelt), so that future repair sites may maximize benefits for species of concern, and
- 2) determine if focus fish species preferentially use Mitigated Repair Sites over unmitigated, unimproved or levee repair sites.

Key results of the 2012 monitoring effort:

- Fish community composition was similar between Mitigated Repair Sites (all design types combined), Naturalized Sites, and Unmitigated Repair Sites, in contrast to findings from the previous year when substantial differences in fish communities were noted between Naturalized, Unmitigated and Repair Sites, as well as among different Repair Types.
- Among Mitigated repair sites, the greatest difference in fish community composition was found between Naturalized Sites, Wetland Trench and No Bench sites, though those differences were minor compared to those found the previous

year (when the largest differences occurred between Unmitigated Repair Sites and 10:1 Slope/Naturalized Sites).

- Juvenile Chinook salmon abundance was higher at Mitigated Repair Sites containing mitigated habitat features (all repair types combined) compared to Unmitigated Repair Sites consisting primarily of rock revetment. This finding is consistent with results from the previous monitoring year.
- Chinook salmon juveniles were more abundant at the 10:1 Slope/Bench Sites than both Naturalized Sites and other Mitigated Repair Site Types. This finding is consistent with results from the previous monitoring year.
- Significantly lower abundance of Chinook salmon juveniles was observed at Wetland Trench Sites compared to both Naturalized Sites and other Mitigated Repair Site Types. This finding is consistent with results from the previous monitoring year.
- Regardless of Mitigated Repair Site Type, juvenile Chinook as well as Chinook salmon fry were more abundant in areas with slow current; this observation is in contrast to findings from 2011, which indicated that both fry and juvenile Chinook salmon were more abundant in areas with increased current velocity.
- Chinook salmon juvenile abundance increased as bank coverage of terrestrial vegetation and terrestrial woody material increased. In contrast, no significant association between these features and juvenile abundance was found the previous year.
- Predation risk to fry and juvenile salmon appears to be highest at Wetland Trench Sites, with elevated predation risk at No Bench Sites. This finding is consistent with results from the previous monitoring year.

1.0 INTRODUCTION AND BACKGROUND

The Sacramento River Bank Protection Project (SRBPP) Long-Term Aquatic Monitoring is part of ongoing work associated with the SRBPP. Authorization & Appropriation for the Program are being provided through: Public Law 86-645 (Flood Control Act of 1960) in accordance with the recommendation of the Chief of Engineers in Senate Document No. 103, 86th Congress, Second Session, "Sacramento River Flood Control Project, California."

The monitoring has been designed to evaluate habitat features of levee repair sites that promote fish use, so that future repair sites can maximize values for fish, and to determine how fish use compares between naturalized sites, various Mitigated repair sites and unmitigated levee repair sites. The Plan was developed in 2008 (H.T. Harvey & Associates et al. 2008) and revised in 2011 (FISHBIO 2011), on behalf of DWR, USACE, NMFS, USFWS, and CFDG to identify monitoring goals and objectives, as well as monitoring protocols to meet the post-construction monitoring requirements of Biological Opinions for emergency levee repairs (see Background). Annual monitoring has been conducted since 2007, but habitat data beginning in 2008 and fisheries data beginning in 2009 is considered the baseline dataset for long-term monitoring evaluations.

This report presents the monitoring results from 2012, with some comparisons to results from previous monitoring conducted in 2009 and 2010 by H.T. Harvey & Associates and PRBO (2011), and in 2011 by FISHBIO (2012). Boat electrofishing was used to document fish presence and estimated index abundance at three different site types: 1) **Mitigated Repair Sites** (i.e., post-2006 emergency levee repair sites that incorporate habitat mitigation features; n = 11 sites), 2) **Unmitigated Repair Sites** (i.e., Unmitigated type of levee repair sites that generally consist of rock revetment without additional habitat enhancement; n=4 sites), and 3) **Naturalized Sites** (i.e., natural bank sites that are not currently have no rock revetment, and are dominated by native riparian or emergent vegetation; n=7 sites). Monitoring at these sites allowed us to compare the efficacy of the habitat improvements incorporated into Mitigated Repair Sites to that of non-mitigated repair designs associated with Unmitigated Repair Sites, as well as to non-engineered conditions at Naturalized Sites.

1.1 Background

On February 24, 2006, Governor Arnold Schwarzenegger declared a State of Emergency for California's levee system, commissioning up to \$500 million of state funds to repair and evaluate State/Federal project levees. Repairs to project levees were conducted under three main programs: the Sacramento River Bank Protection Project (SRBPP), the PL84-99 Rehabilitation Program, and the former Sacramento-San Joaquin Erosion Repairs Project (SSJERP). The USACE is the lead agency for the SRBPP and the PL 84-99 Rehabilitation Program. The California State Department of Water Resources (DWR) was the lead agency for the SSJERP but is currently being replaced by the Flood System Repair Project, which will address deficiencies in the entire State Plan of Flood Control as designated in the State of California Central Valley Flood Protection Plan. .

Environmental mitigation measures were not required during the initial SRBPP First Phase erosion repairs. Since enacting of National Environmental Policy Act and California Environmental Quality Act in 1970 mitigation habitat measures were slowly incorporated into levee design. Habitat protection and creation of on-site riparian habitat features were not incorporated until the early 1990s listing of under state and federal Endangered Species Acts for winter-run (SE 1989, FE 1994) and spring-run (SE 1999, FT 1999) Chinook salmon (*Oncorhynchus tshawytscha*), steelhead (FT 1998) (*O. mykiss*), and the delta smelt (ST 1993, SE 2010, FT 1993) (*Hypomesus transpacificus*) in the Sacramento River . As a result, mitigation measures for channel and river bank modifications were developed for the post-2006 levee repairs and were designed based on the Standardized Assessment Methodology (SAM; USACE 2004, 2006; USACE and CVFPB 2009). The SAM is a predictive model used to quantify potential impacts to threatened and endangered fish species due to construction of bank revetment and other bank protection measures, and it can also be utilized to assess the fish habitat value of experimental mitigation and compensation measures. The SAM quantifies habitat values in terms of a bank line or area-weighted species response index; habitat variables include bank slope, floodplain inundation ratio, bank substrate size, instream structure, aquatic vegetation, and overhanging shade.

Four typical levee repair designs were developed and constructed in the project area. Recently completed (post-2006) emergency levee repairs (Mitigated Repair Sites) include constructed habitat features to offset the impacts from loss of riparian habitat due to the levee repair action. Purposes of constructed habitat features include: 1) protecting the toe and upper slopes of the bank with rock revetment and/or rock revetment mixed with soil; 2) establishing a berm around the mean summer water surface level to provide aquatic habitat during lower and higher river stages in winter and spring; 3) placing instream woody material (IWM) to enhance fish habitat value; and 4) planting live pole cuttings, container plantings and grasses to stabilize the bank and provide riparian and shaded riverine aquatic habitat features. Rock revetment provides bank and levee protection, while the other habitat features provides benefits to fish and other aquatic and riparian organisms.

The National Marine Fisheries Service (NMFS) and U.S. Fish and Wildlife Service (USFWS), issued of Biological Opinions (Opinions) for critical levee erosion repair effects based on ESA Section 7 consultations. (NMFS 2006a, 2006b, 2006 c, 2007, 2008, USFWS 2006a, 2006b, 2006c, 2007). The NMFS and USFWS 2006-2007 Opinions supported approval of critical levee repairs at 54 sites (along the Sacramento and Bear rivers; and Steamboat, Sutter, and Cache sloughs). DWR undertook repairs at 27 of the sites, and the USACE undertook repairs at the remaining 27 sites. These sites were identified as “critical,” as defined by bank erosion that could threaten the structural integrity of the Sacramento River flood control system, and therefore required immediate work to prevent levee failure during the next flood (Ayres Associates 2007).

The terms and conditions of these critical erosions sites Opinions and SRBPP remaining 24,000 linear feet in the second phase (NMFS 2008b) required long-term monitoring of habitat features and fish at SRBPP sites for the purpose of characterizing the success of the habitat features in replacing the natural features at the site. In accordance with these Opinions, the USACE used the SAM, in coordination with USFWS, NMFS, cooperating state agencies and consultants to quantify potential impacts to focus species (i.e., threatened, endangered fish, and candidate

species under the federal ESA and fish species with Essential Fish Habitat designations under the Magnuson-Stevens Fisheries Act) from “bank revetment and other bank protection projects as well as experimental mitigation and compensation measures that may be implemented within the SRBPP planning area” (USACE 2004, 2006; USACE and CVFPB 2009). Focus species and potential predators of the focus fish species addressed in the SAM include:

- Sacramento River winter-run Chinook salmon (focus species; federally Endangered)
- Central Valley spring-run Chinook salmon (focus species; federally Threatened)
- Central Valley fall/late fall-run Chinook salmon (focus species; federally Species of Concern)
- Central Valley steelhead (focus species; federally Threatened)
- Delta smelt (Focus species; federally Threatened)
- Sacramento pikeminnow (native predator)
- Inland silverside (introduced predator of delta smelt eggs/larvae only)
- Striped bass (introduced predator)
- Black bass (largemouth and smallmouth; introduced predators)

In June 2008, H.T. Harvey & Associates (2008) prepared the initial fish and habitat monitoring plan on behalf of DWR, USACE, NMFS, USFWS, and CDFG to meet the monitoring goals and objectives determined by the Interagency Flood Management Collaborative Program Critical Repair Technical Task Sub-Group (8 February 2008). The monitoring goals and objectives were formulated to meet NMFS and USFWS Opinion requirements that the data evaluate whether “the various experimental ... on-site compensations are functioning in a manner that enhances habitat value and offsets adverse bank protection effects” (NMFS 2001, 2004, 2006a, 2006b, 2006c, 2007; USFWS 2004a, 2004b, 2006a, 2006b, 2006c, 2007). A revised plan was developed and implemented by FISHBIO in 2011, with revision pertaining mainly to monitoring site selection (sites were chosen to minimize incidental take of delta smelt and to provide a monitoring design with balanced replicates for each of the repair types).

2.0 STUDY AREA DESCRIPTION

The Sacramento River originates on the slopes of Mount Eddy in the Trinity Mountains within the Klamath Mountain Range (Thompson 1961). It joins with the McCloud and Pit rivers at Lake Shasta to form the largest river in California, with a drainage basin approximately 335 miles (540 km) long and 168 miles (270 km) wide (Larsen and Greco 2002). The Sacramento River basin climate is Mediterranean with mean annual rainfall of 36 inches (914 mm) and average temperatures ranging from 54°F (12°C) in January to 88°F (31°C) in July (Domagalski et al. 2000). The Sacramento River system flows are regulated by dams built for water supply and flood control, which have stabilized and altered flow regimes, changing the river's erosion and sediment transport characteristics (Mount 1995).

The SAM planning area within the Sacramento River mainstem extends from the lower Sacramento River near Collinsville at RM 0 to Chico Landing at RM 194. The SAM planning area also includes tributaries, distributaries, and bypasses of the Sacramento River Flood Control Project but not included in the SRBPP fish monitoring plan. Based on differences in reach geomorphology and vegetation as described below (and in further detail in Stillwater Sciences and Dean Ryan Consultants 2004), four reach designations of the Sacramento River are identified in the SAM:

- Reach 1a: from Collinsville to (near) Isleton (RM 0-20)
- Reach 1b: from (near) Isleton to Verona (RM 20–80)
- Reach 2: from Verona to Colusa (RM 80–143)
- Reach 3: from Colusa to Chico Landing (RM 143-194)

The lower reaches of the Sacramento River (1a and 1b) along RM 0-80 run past the City of Sacramento to the Delta (Stillwater Sciences and Dean Ryan Consultants 2004). Here, the majority of the river is narrowly confined by large levees. The riparian vegetation tends to consist of discontinuous narrow bands of riparian vegetation along the levees. The soils along these reaches are primarily composed of Colombia and Sycamore series (SRAC 2003). A portion of the reach (RM 0 to approx. RM 30) is tidally influenced.

The middle reach of the Sacramento River extends from RM 80, near the Feather River confluence, north towards RM 143 near Colusa (Reach 2; Stillwater Sciences and Dean Ryan Consultants 2004). The channel is generally wider than in the lower reaches; however the river is confined within levees.

The levees along the uppermost reach from RM 143-194 (Reach 3) are generally set back and spaced wide enough to allow for channel migration (Stillwater Sciences and Dean Ryan Consultants 2004). In areas not fortified with rock revetment or other erosion control features, the meandering river forms cut banks, gravel bars, and associated floodplains. Sparse remnant stands of native riparian forest occur along the upper reach and early successional species are established on alluvial deposits.

Numerous critical erosion repairs site were constructed in the SAM planning area since the State of Emergency for California's levee system was declared in 2006. The majority of levee repairs

occurred in Reaches 1a and 1b, with fewer in Reaches 2 and 3. Sites included in the monitoring effort described herein include levee repair sites (both Unmitigated Repair Sites and post-2006 Mitigated Repair Sites) and Naturalized Sites between RM 24 and 91.9 on the Sacramento River (i.e., located within SAM Reaches 1b and 2).

Critical erosion repairs included several different habitat features that vary in construction cost, are constrained by site-specific conditions (e.g., hydraulics) and were hypothesized to provide habitat more suitable than rock revetment to fish species of interest. SRBPP monitoring sites include representative conditions at four (4) different erosion repair design types (Figures 1-4). They are:

- 1) **10:1 Slope design.** Sites with the 10:1 Slope design tend to have more gradual slopes (10:1) than the other design types;
- 2) **Bench design.** The Bench Sites have a riparian or emergent bench designed to be inundated during periods of increased flows or high tide events, depending on location;
- 3) **Wetland Trench design.** Wetland Trench (a.k.a., Dietl ditch) sites have an off-channel ditch, parallel to the main channel, designed to provide shallow water aquatic habitat, generally with dense emergent vegetation. These trenches are connected to the main channel and inundate on a regular basis;
- 4) **No Bench design.** No Bench Sites typically have steeper slopes (2:1 or 3:1) and do not include a riparian bench.

Mitigated Repair Sites were selected to exemplify conditions at the four design types and include a number of variables including SAM reach designations, geomorphic setting, percent vegetation, and bank slope. Diversity of habitat features among the available Mitigated Repair Sites is lower than that found within the study area overall; therefore, Unmitigated Repair and Naturalized Sites were selected with the intent of maximizing the range of habitat variables represented. For the purposes of this report, Unmitigated Repair Sites refer to locations where engineered banks consist largely or exclusively of rock revetment and do not include on-site mitigation features designed to improve aquatic habitat. Naturalized Sites refer to locations where the shoreline consists of naturally deposited sediment, does not contain rock revetment and are dominated by native riparian or emergent vegetation. Additional details regarding site selection strategies are provided in H.T. Harvey & Associates and PRBO (2011).

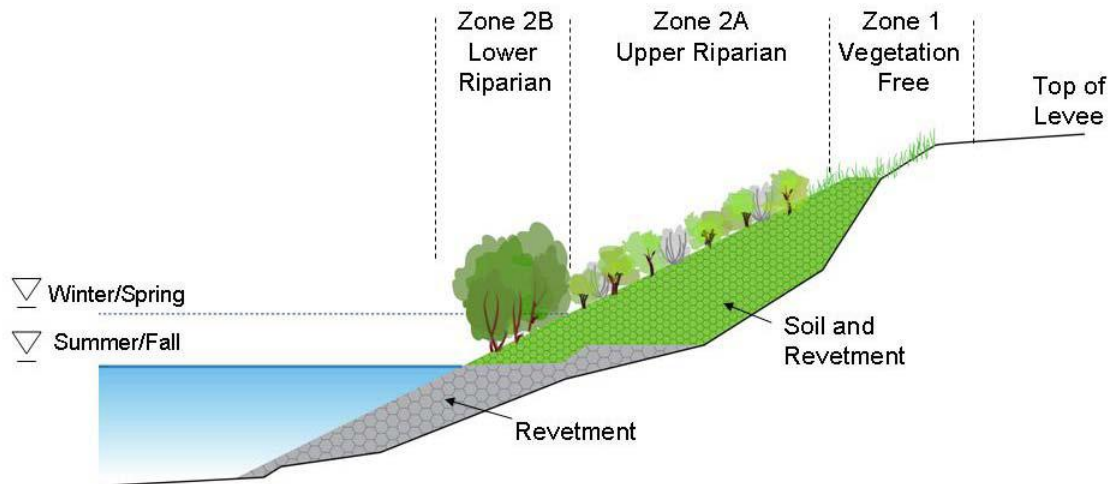


Figure 1. Bank fill rock slope with revegetation (Source: USACE 2007b).

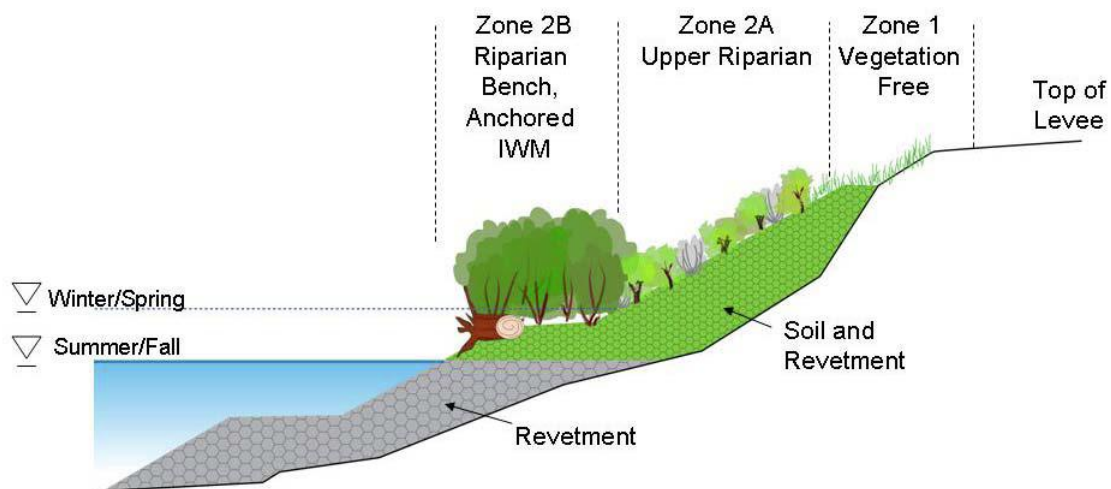


Figure 2. Low riparian bench with revegetation and anchored in-stream woody material enhancements *above* the summer/fall waterline (Source: USACE 2007b).

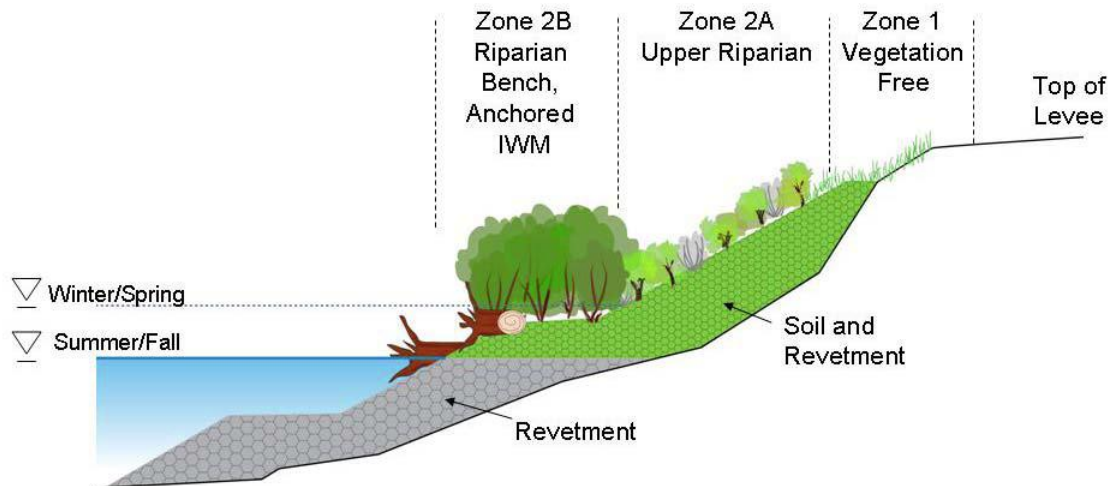


Figure 3. Low riparian bench with revegetation and anchored in-stream woody material enhancements *above and below* the summer/fall waterline (Source: USACE 2007b).

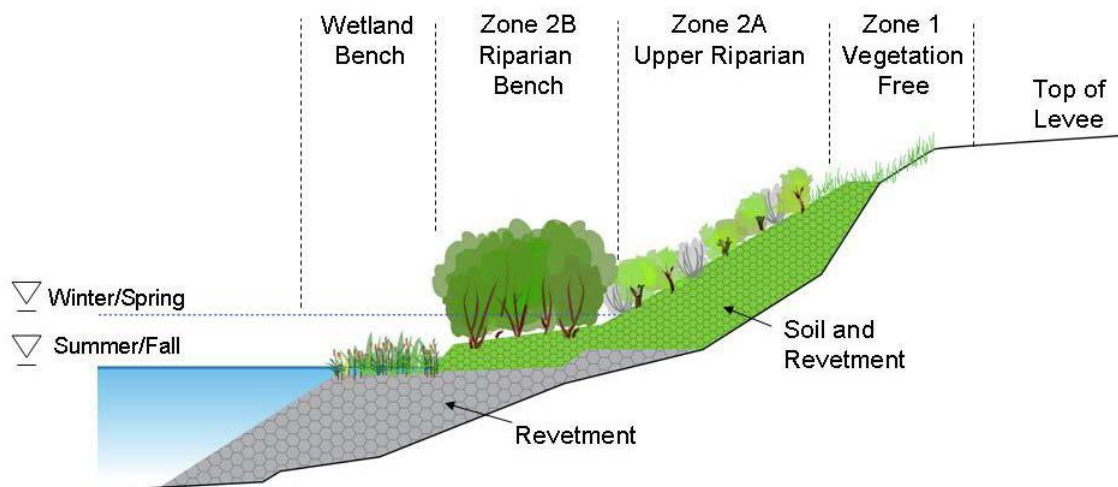


Figure 4. Low riparian and wetland benches with revegetation in tidally influenced areas (Source: USACE 2007b).

3.0 MONITORING OBJECTIVES

Specific goals and objectives have differed slightly between years. During initial monitoring for SRBPP sites in 2007, Stillwater Sciences (2009) identified two objectives and stated that results of monitoring could be “used to improve designs of future bank protection projects, to better inform offsite habitat compensation needs, and to update the SAM model (USACE 2004).” Objectives included:

Objective 1. Compare fish abundance and species composition at bank protection sites and natural bank areas, and

Objective 2. Evaluate the relationship between fish abundance and the different constructed habitat features.

In 2008, objectives were further refined and specific goals were identified for the Long-Term Aquatic Monitoring (H.T. Harvey & Associates et al. 2008). These goals and several of the refined objectives were implemented in 2011, as follows:

Goal 1. Determine if on-site mitigation at mitigated Repair Sites is approximating ‘Naturalized banks’¹, and

Goal 2. Identify which habitat features of mitigated levee repair sites improve habitat for focal fish species, namely anadromous salmonids, as indicated by increased abundance in relation to these features

Fish Objective 1. Determine the fish species present at the constructed habitat features and levee benches at various seasonal flow levels at the bank protection sites after site construction and at other reference sites identified for monitoring (i.e., Naturalized and Unmitigated Repair).

Fish Objective 2. Evaluate fish community abundance and composition using electrofishing at selected Naturalized and Repaired (mitigated and non-mitigated) Sites. Information from these surveys will be used to determine salmonid presence and abundance (use) at sites and microhabitats within sites, as well as the relative risk of predation to juvenile salmonids at these sites.

Habitat Objective 1. Characterize the habitat features located at various seasonal water elevations at selected bank protection and reference sites. These data will be collected using a repeatable survey design and in a format that can be used to model focus fish species response per the SAM model requirements. Each selected site will be monitored with photography using two to six permanent photo points established at each site. Alternatively, a panoramic method for photographic recording of each site may be implemented.

¹ “Naturalized banks” are those river banks that have not been engineered and are not riprapped; they are dominated by native, naturally established vegetation.

4.0 METHODS

Fish abundance and habitat monitoring were conducted in spring 2012 according to the objectives of the Program. Unlike previous years, telemetry was not conducted as part of this report. Survey methodologies have evolved and are continuously improved to adapt to unforeseen challenges and incorporate insights gained in this ongoing monitoring and evaluation effort.

4.1 Fisheries Monitoring (Fish Objectives 1 and 2)

Boat electrofishing is one of the few effective sampling methods for large rivers and was used for sampling areas upstream of RM 20 to meet Fish Objectives 1 and 2. SRBPP project sites downstream of RM 20 were restricted from using any type of sampling method (i.e., electrofishing or trapping) to avoid take of delta smelt. Sites were electrofished using an SR-18 electrofishing boat (Smith-Root, Inc., Vancouver, Washington). Sampling occurred on three events in 2012 (March 17-20, April 2-4, and April 16-18) to encompass outmigration timing of fall-run, late fall-run, spring-run, and winter-run Chinook salmon and steelhead (Snider and Titus 2000, Williams 2006). Sampling events were initiated in response to increases in turbidity at Knight's Landing Rotary Screw Trap and Sacramento River flows below Wilkins's Slough (WLK, RM 117.6, CDEC).

The number and location of electrofishing monitoring sites has varied since 2007 (Table 1). Similar to previous years, the study sites were categorized into Regions based on location and were identified as Knights Landing (RM 71 to 95), Garcia Bend (RM 40 to 70), and Ko-ket (RM 21 to 39). All 22 sites were sampled during each of three sampling events in 2012 (Table 1 and Figure 5). Sixteen of the sites had been included in initial assessments (2008-2010) by H.T. Harvey & Associates and PRBO (2009, 2010, 2011); while the remaining sites were first sampled in 2011 (FISHBIO 2012). Sites added to the monitoring program in 2011 included one Undulating Bench Site (RM 72.2R), one Setback Site (RM 57.2) and five Unmitigated Repair Sites (i.e., pre-2006 unmitigated levee repair sites; RM 24.0R, RM 38.7R, RM 56.0R, RM 85.6L, and RM 90.8L, Table 1). In 2012, a 10:1 Slope repair site (RM 87.0L) was added and the single setback levee site (RM 57.2R) was removed since there were no other similar design types and there was a high likelihood of capturing adult salmon at this site as a result of adjacent channel configuration. Sampling sites ranged from approximately 352 to 709 feet (107 to 216 meters) in length.

The sampling approach was similar to 2011, which was designed to sample a greater proportion of each site than sampling conducted prior to 2011. In lieu of the previous grid point sampling method (conducted in 2009 and 2010), each sampling site was considered a unit and was sampled in its entirety. At each site, sampling was initiated by approaching the shoreline with activated electrodes for about 10-15 seconds, then turning off the electrodes while maneuvering away from the shoreline. This was repeated multiple times, beginning at the downstream boundary of the unit and continuing with repeated approaches until the upstream boundary of the unit was reached. Sampling was conducted as close as possible to the shoreline and each approach to the bank was designated as a sampling point. The number of points sampled at each site varied between 22 and 57, depending on site and sampling event (Table 1).

Table 1. Location and characteristics of electrofishing sampling sites number of points sampled at each site (ratio of site length to number of points in each sampling event) and repair design status (activated or not during sampling) for 2012 monitoring.

Site (RM) ¹	SAM Reach	Region	Site Type	Sample Unit Length (feet)	2007 and/or 2008	2009 and 2010	2011	2012					
								Number of points sampled (feet/point)			Repair Design Activated (Y/N/P)		
								F1 ²	F2 ²	F3 ²	F1	F2	F3
91.9R	2	Knights Landing	Naturalized	586		X	X	22 (26.6)	38 (15.4)	36 (16.3)	-	-	-
90.8R			Naturalized	553		X	X	30 (18.4)	34 (16.3)	39 (14.2)	-	-	-
90.8L			Unmitigated	637			X	37 (17.2)	28 (22.8)	43 (14.8)	-	-	-
87.0L			10:1 Slope	602				38 (15.8)	49 (12.3)	36 (16.7)	Y	Y	Y
85.6R			No Bench	536		X	X	37 (14.5)	27 (19.9)	39 (13.7)	Y	Y	Y
85.6L			Unmitigated	518			X	37 (14)	47 (11.0)	43 (12.0)	-	-	-
72.2R	1B		Bench	657	X		X	48 (13.7)	51 (12.9)	41 (16.0)	Y	Y	Y
69.9R	1B	Garcia Bend	No Bench	644		X	X	71 (9.1)	42 (15.3)	43 (15.0)	Y	P	P
69.1L			Naturalized	610		X	X	52 (11.7)	43 (14.2)	47 (13.0)	-	-	-
57.2R			Setback	528			X	NS	NS	NS	NS	NS	NS
56.0R			Unmitigated	581			X	31 (18.7)	56 (10.4)	30 (19.4)	-	-	-
50.3R			Naturalized	352		X	X	24 (14.7)	31 (11.4)	24 (14.7)	-	-	-
50.2L			No Bench	652	X	X	X	39 (16.7)	44 (14.8)	34 (19.2)	Y	P	N
47.9R			10:1 Slope	592	X	X	X	47 (12.6)	41 (14.4)	28 (21.1)	Y	P	P
47.0L			10:1 Slope	669	X	X	X	35 (19.1)	42 (15.9)	39 (17.2)	P	N	N
44.7R			10:1 Slope	709	X	X	X	40 (17.7)	54 (13.1)	47 (15.1)	P	N	N
43.7R			10:1 Slope	552	X	X	X	48 (11.5)	54 (10.2)	39 (14.2)	P	N	N
38.7R	1B	Ko-ket	Unmitigated	579			X	39 (14.8)	41 (14.1)	41 (14.1)	-	-	-
34.5R			Wetland Tr.	585	X	X	X	40 (14.6)	40 (14.6)	37 (15.8)	Y ³	P ³	P ³
26.9L			Wetland Tr.	463	X	X	X	36 (12.9)	36 (12.9)	35 (13.2)	N ³	N ³	N ³
25.0L			Naturalized	652	X	X	X	56 (11.6)	49 (13.3)	54 (12.1)	-	-	-
24.0R			Unmitigated	677		X	X	57 (11.9)	49 (13.8)	48 (14.1)	-	-	-
24.0L			Naturalized	693		X	X	50 (13.9)	42 (16.5)	47 (14.7)	-	-	-

¹ Individual sites are identified by river mile and left/right designation; ² Designations correspond to the three sampling events in 2012; ³ sites are tidally influenced and activation status reflects inundation at time of sampling.

Codes: Unmitigated= rock revetment levee repair; NS =not sampled; NV=no video; Y=yes, N=no, P=partial.

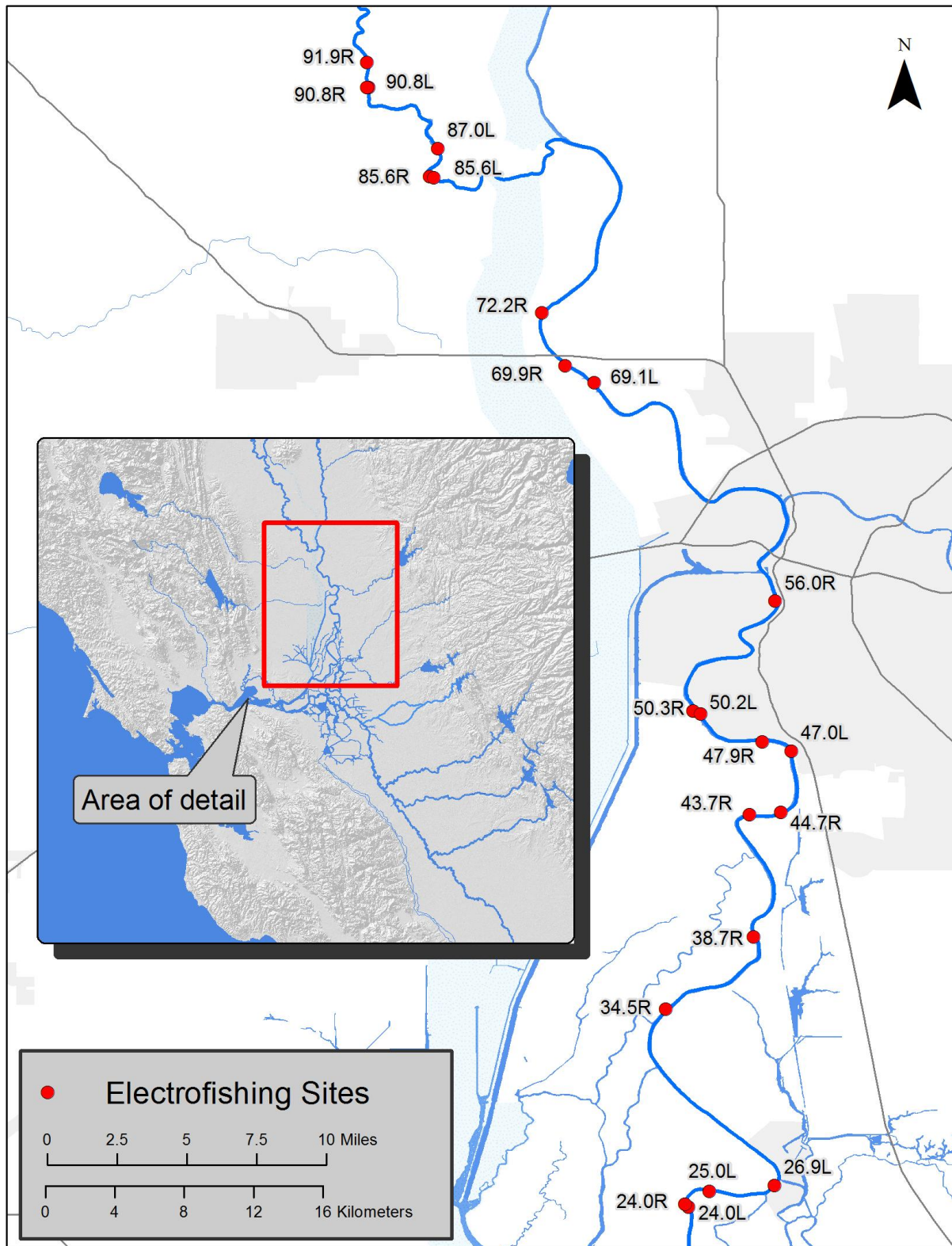


Figure 5. Location of 2012 monitoring sites. Note that site 87.0L was added as a new site in 2012, while sampling of site 57.2R (setback levee) was discontinued.

While at some sites the design features intended to enhance fish habitat were either partially or not activated (i.e., inundated) at the time of sampling, sampling occurred regardless of inundation level (Table 1). A partially activated site indicates that some, but not all, design features were inundated. For example, during the first sampling event, woody material placed at RM 52.0L was inundated during the first sampling event, yet some of these features were above the wetted margin during the second sampling event, and almost exclusively out of the water during the third event. Design activation levels were determined through review and comparison of electrofishing GPS tracks and site photos taken during each electrofishing event with photos taken during SAM summer habitat surveys. For Wetland Trench Sites, the entire margin (outside of trench) was always sampled and sampling within the trench was conducted whenever possible; however, dense vegetation and shallow water sometimes limited sampling to the inlets and outlets.

At each sampling site, voltage, pulse rate, and amperage ranged from low (50-500) to high voltage (100-1000), 30-60 pulses/s, and duty cycle of 20%-60%, respectively; starting voltages, pulse rates, and amperages were determined by final settings at the previous sampling site and adjustments were made based on electrical conductivity or fish response. During electrofishing, a continuous track of the boat movement was recorded via GPS (Trimble; Geo XH 2005), and a video camera (with audio) was used to document activities. Sampling start and stop times were recorded.

Personnel stationed at the front of the boat captured stunned fish using dip nets. Missed fish (i.e., fish that were observed but escaped the net) were noted, with fish identified to species and approximate size recorded whenever possible. Fish collected were placed in an aerated livebox until processing (i.e., measuring and recording) was complete. A partition in the livebox allowed separation of potential predators from prey-sized fish to eliminate harmful interactions. Sampling was conducted continuously, and all captured fish were processed simultaneously following completion of sampling at each unit.

Fish Processing

During processing, all fish captured were identified to species when possible and measured to the nearest millimeter (total length, or fork length for species with forked tails, such as salmon and steelhead). Whenever large numbers of an individual species were captured during a single sampling event in a particular sample site, all fish were counted and identified, but measurements were only collected from a random sub-sample of 50 individuals (for salmon and *O. mykiss*) and 20 individuals per species for non-salmonids. Sub-samples of salmonids were stratified by lifestage, with 50 measurements collected within each lifestage (i.e., fry <50 mm, parr=50-69 mm, smolts $\geq$ 70 mm for Chinook salmon; and fry <100 mm, juveniles 100-299 mm, and adults >300 mm for steelhead). Condition of all fish was rated as “alive”, “injured”, or “mortality”, and smolt index was recorded for all salmonids (seven category scale, where 1 = yolk-sac fry, 2 = fry, 3 = parr, 4 = silvery parr, 5 = smolt, 6 = mature adult, and IAD = immature adult; Interagency Ecological Program, unpublished). Sensitive species/lifestages were anesthetized prior to processing using a solution of Tricaine methanesulfonate (Western Chemical, Inc.; Tricaine-S) at a 26.4 mg/L concentration, and were allowed to recover prior to release at the upstream boundary of the sampling unit.

Physical Habitat Characteristics

In 2012, various environmental and physical parameters were recorded within each site to document the general physical characteristics present at each site during the electrofishing survey. Prior to electrofishing each site, environmental conditions were recorded immediately adjacent to the site in the mid-channel. After electrofishing each site, physical parameters were measured along five evenly spaced transects oriented perpendicular to the shoreline to adequately capture the variability at each site. At each transect, depth and mean column velocity were measured, and substrate type (sand/silt, gravel, or revetment) was recorded at 5, 10, and 15 feet (from the bank to reflect the distance from shore where electrofishing was initiated during each approach. Depth and velocity measurements were averaged to estimate mean depth and velocity for each sampled unit. Measurement of environmental and habitat parameters included the following:

- Water temperature and dissolved oxygen: measured in Celsius and mg/L, respectively, with an optical dissolved oxygen meter (YSI; ProODO).
- Conductivity and salinity: measured in $\mu\text{S}/\text{cm}$ and ppm, respectively, with an Extech EC400 multi-range sensor.
- Water clarity: measured with a Secchi (depth in feet; LaMotte Company); and by water sample collected and analyzed immediately using a turbidimeter (Oakton; model T-100). Water samples were collected prior to electrofishing adjacent to site towards mid-channel to ensure samples were not biased due to substrate disturbance caused by boat wakes or other wave activity.
- Water depth: measured in feet with a marked polyvinyl chloride (PVC) rod.
- Mean column water velocity: measured in meters per second (m/s) with a digital flow probe (Marsh-McBirney or Global Water Instrumentation) at 0.6 of total depth in locations less than 2.5 feet (approx. 0.75 m); and average of readings at 0.2 and 0.8 of depth in locations 2.5 feet deep or more.
- Substrate type was characterized as sand/silt or rock by probing the substrate with a PVC rod, and the presence of submerged vegetation was recorded for each transect point.
- The location of shoreline (water's edge) was recorded using a sub-meter GPS.

Fish Presence/Absence and Habitat Features at Sampling Points

Video/audio recordings and written notes were reviewed to determine fish presence and abundance (actual catch plus missed fish identifiable to species), and habitat features at each sampling point, including

- Density of emergent woody material and/or emergent vegetation² (none, low, medium, high);
- Percent coverage of terrestrial vegetation and/or terrestrial woody material (visually estimated during video review), considered informative to habitat condition during inundation at higher flows;
- Absence/presence of shade;
- substrate (rock revetment versus mud/silt/sand).

² Emergent vegetation was defined as any live, elastic plant

4.2 Sampling with Alternative Collection Methods

To investigate the applicability of alternative collection techniques to the Sacramento River Bank Protection Project Long-Term Aquatic Monitoring Program, FISHBIO deployed a prepositioned area electrofishing grid, Breeder traps, and quatrefoil light traps on June 20, 2012. Sampling was conducted upstream of Knights Landing, CA, within or in the vicinity of monitoring sites at RM 90.8R and RM 91.9R.

Prepositioned Area Electrofishing

Prepositioned area electrofishing (PPAE) is a sampling technique that was developed for sampling fish without introducing behavioral bias by scattering fish upon approach (see Bain et al. 1985). A rectangular electrode frame, consisting of rope intertwined with copper wire (electrodes) and nylon rope to ensure appropriate spacing (which depends on the conductivity at the site and the voltage output), is anchored to the riverbed (Figure 6). The site is then vacated for a minimum of 15 minutes to allow fish to return to their natural distribution.

This method allows for sampling of specific microhabitats, as electrodes can be placed to target habitat features of interest. It allows for accurate sampling of fish communities as behavioral bias is largely eliminated. Also, the sampled area is easily quantifiable, as electrode lengths and spacing are known. While this method is very effective for sampling specific habitat units, obtaining large sample sizes is difficult due to the time it takes to install the sampling equipment and the required resting period following installation. Using several electrode frames simultaneously may alleviate this problem.

Two PPAE grids of slightly different design (electrodes constructed of copper wire or stainless steel cable) were deployed simultaneously in at one of the sites specified above to evaluate microhabitat usage of focal species (salmonids and piscivorous species); units were deployed in several shallow water habitat and connected via an extension cord (50 feet/15 meters long) to a boat-based generator (while this technique can also be employed using a shore based generator, the boat based approach was preferred since it increased mobility in areas with limited shore access). After deploying the grids, the area was left undisturbed for approximately 20 minutes to allow fish to resume their natural behavior. Following this period, netters then carefully approached the sampling locations and the grids were activated for up to 10 seconds. If applicable, stunned fish were netted and transferred to a bucket for identification and recovery. After completion of fish processing and contingent upon satisfactory performance of sampling equipment (immobilization of target species with no or minimal injury/mortality), units were removed from the water and re-deployed at nearby habitat features to obtain a minimum of three replicates.

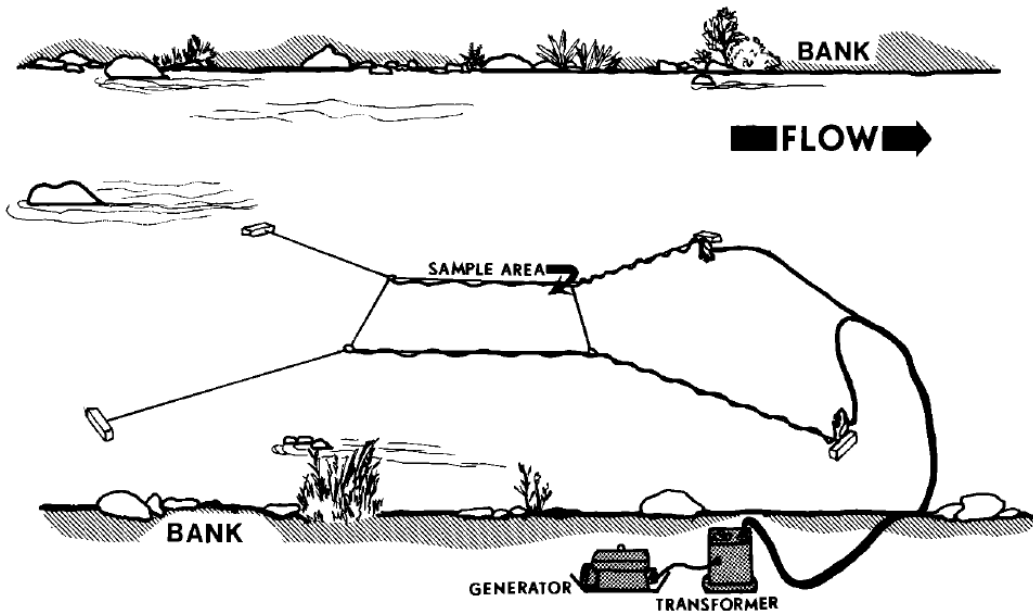


Figure 6. Schematic of prepositioned area electrofishing (from Bain et al. 1985).

Breder traps

Breder traps are designed to target larval and juvenile fish and are constructed of transparent materials, which may allow fish to be identified, enumerated, assigned a size class or life stage, and released without direct handling. This type of trap allows fish to freely swim within the trap; however, due to the orientation of the entrance, fish have difficulty exiting the trap. When removed from the water, the Breder trap holds water, which allows fish to be transported to release location without netting or any further handling, if desired. Fish can also be poured into buckets to allow for more detailed processing (such as obtaining length measurements), which minimizes handling associated with net transfers.

Breder traps have been used to successfully target juvenile and larval fish species (Benbow et al. 2004; Sargent and Carlson 1987). Breder traps have a simple design (Figure 7) and are relatively easy and inexpensive to construct from Plexiglas or similar material. The small size of Breder traps allows them to be deployed in areas that are difficult to sample using other methods (e.g. among instream woody material and vegetation). Deployment of several traps simultaneously at restoration and control sites may be able to provide spatial and temporal resolution of fish densities in specific microhabitats (Sargent and Carlson 1987). As with similar trap designs (e.g., light traps), Breder traps can be modified to select for specific fish sizes by adjusting the size of the opening. The most commonly used configuration of 15 cm x 30 cm with a 12 mm opening has been found to capture fish up to 60 mm. Yellow chemical lights (calyme light sticks) have also been found to result in increased catch by attracting fish to the traps under turbid conditions (Gerhke 1994; Humphries et al. 2002) and were used in conjunction with Breder traps in this survey.

Two Breder traps were “baited” with chemical light sticks (Coleman Illumistick® glow sticks, orange and yellow, respectively; Figure 17), and deployed in shallow water (1.5 and 2.7 feet/

0.45 and 0.82 m) near dense accumulations of small diameter woody material or overhanging vegetation, with the trap opening facing downstream. Both traps were fished for two 45 minute intervals prior to (7:30 pm – 9:00 pm) and after nightfall (9:00 pm – 10:30 pm, a total of 8 trapping intervals).

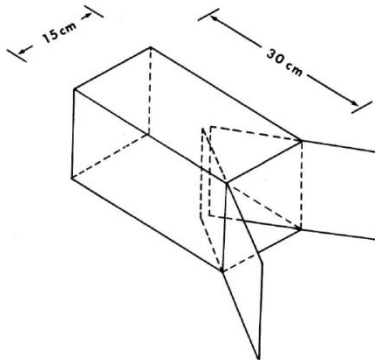


Figure 7. Breder trap schematic (from Sargent and Carlson 1987).

Quatrefoil light traps

Quatrefoil light traps are similar in size and concept to Breder traps, but differ in that they have four vertical openings that may increase trap efficiency (Figure 8). Vertical openings are created by mounting four 15-30 cm long Plexiglas cylinders (10-15 cm diameter) to a 30.5 X 30.5 cm Plexiglas sheet (Figure 8). This trap design also incorporates an (optional) artificial light source to attract fish in turbid water. Similar to Breder traps, light traps can be deployed in areas that are difficult to sample using conventional techniques.

A foam plate (with a reflective underside to increase light intensity) was glued to the top of the quatrefoil light traps to provide flotation so the trap would attract small fish near the water's surface. A collection container with a fine mesh screen was connected to the bottom of the trap, designed to collect captured individuals when the trap is removed from the water.

Two traps were “baited” with chemical light sticks (Coleman Illumistick® glow sticks , two orange and two yellow light sticks, respectively), and deployed simultaneously in relatively shallow water (2.8 and 3.5 feet/ 0.85 and 1.07 m) near dense accumulations of small diameter woody material or overhanging vegetation. Both traps were fished for two 45 minute intervals prior to (7:30 pm – 9:00pm) and after nightfall (9:00 pm – 10:30 pm, a total of 8 trapping intervals). A fine mesh net was placed around the traps prior to retrieval to capture any fish that may escape through the trap openings as water drains from the trap during trap retrieval.

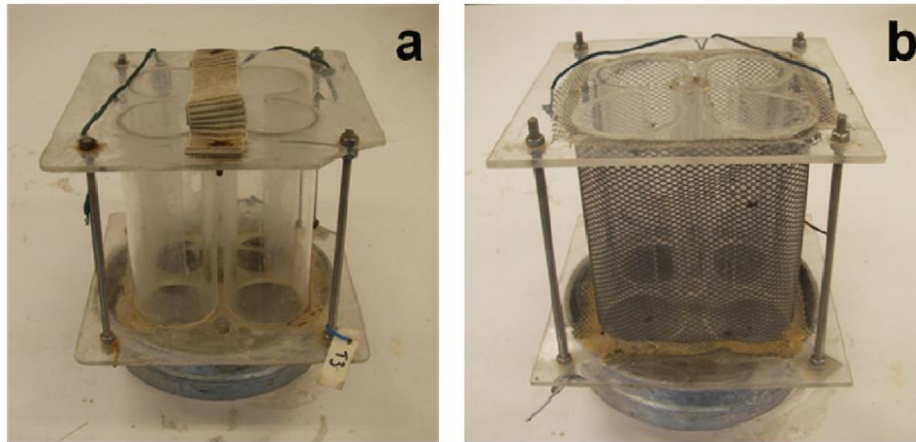


Figure 8. Traditional Quatrefoil light trap (a), and trap modified with selective mesh (b).

4.3 Habitat Monitoring (Habitat Objective 1)

SAM habitat monitoring efforts were conducted by the USACE in 2012 (in prep.).

FISBIO employed photo documentation and side-scan sonar at the SRBPP long term monitoring sites to meet Habitat Objective 1: to document habitat conditions at each site for comparison to previous surveys in order to evaluate the evolving conditions at each site.

4.3.1 Photo Documentation

Photo documentation occurred during electrofishing surveys in March and April in 2012. A 12.1 megapixel/6X zoom digital camera (Canon PowerShot G9) was used to take several overlapping images that encompassed the entirety of each site (images taken from the boat while positioned approximately mid-channel). Whenever possible, images were stitched together to produce a semi-seamless panoramic of individual sites. Steep banks and dense vegetation encountered at many SRBPP areas made traditional photo point monitoring impractical at many sites. Furthermore, channel widths encountered were typically too wide to permit establishment of photo points on opposing banks due to equipment focal limitations. It was determined that photos taken from the boat, mid-channel, would better document site conditions under most circumstances. At some sites, the orientation of the sun created glare that necessitated boat positioning in a manner different than previously described, generally closer towards either the upstream or downstream end of the project site.

4.3.2 Sidescan Sonar

Habitat features encountered by fish are potentially underrepresented, or misrepresented, using traditional Standard Assessment Methodology (SAM; Corps 2004) habitat surveys conducted during low flow months (summer/fall). Side scan sonar (SSS) technology has been used to provide quantitative assessments of instream woody material otherwise hidden by deep turbid water (Kaesler and Litts 2010). By collecting habitat data during electrofishing surveys using

side scan sonar (SSS), georeferenced images of underwater features can be used to augment/compliment the SAM habitat surveys and to potentially identify temporal changes in habitat structure.

Underwater sonar imagery of project sites was collected immediately before or after electrofishing surveys using a boat mounted side imaging sonar/GPS unit (Humminbird, Eufaula, AL; 1197c). Concurrently, boat tracking data was collected with a sub-meter GPS (Trimble, Sunnyvale, CA; Geo 2005). The resulting imagery was processed using methods described in Kaeser and Litts (2010) to produce a single seamless mosaic of submerged areas within and immediately adjacent to project sites. Imagery was anticipated to document existing IWM (≥ 10 cm diameter and ≥ 1.5 m in length) and substrate to help evaluate habitat use by fish species collected during electrofishing.

5.0 DATA REPORTING AND ANALYSIS

5.1 Fisheries Monitoring (Fish Objectives 1 and 2)

Steelhead were not captured in sufficient numbers (n=23) during 2012 Long-Term Aquatic Monitoring to warrant their inclusion during analysis; therefore, only Chinook salmon catch data were used for evaluating habitat use by salmonids.

5.1.1 Species Captured (Fish Objective 1) and Fish Community Abundance and Composition (Fish Objective 2)

In order to determine the fish species present at selected Naturalized and Repair Sites, the total number of different fish species recorded (captured and missed) were identified. Size (length) distributions of captured focus fish and predatory species were examined via histograms.

In order to evaluate fish community abundance and composition at selected natural and repaired sites, community composition at four different types of Mitigated Repair Sites (10:1 Slope, Bench, No Bench, and Wetland Trench), Naturalized Sites, and Unmitigated Repair Sites (i.e., six site types) were recorded and quantified. Similarities between community composition at these six different sites were examined for all pair-wise comparisons with Morisita's Index (Morisita 1959, as revised by Horn 1966), and calculated as

$$C_{\lambda} = \frac{2\Sigma(X_{ij} * X_{ik})}{(\lambda_j + \lambda_k)(\Sigma X_{ij} * \Sigma X_{ik})}$$

Where C_{λ} is the index of similarity between site type j and k , X is the number of individuals of species i at site type j and k , and

$$\lambda_j = \frac{\Sigma X_{ij}^2}{(\Sigma X_{ij})^2} \quad \lambda_k = \frac{\Sigma X_{ik}^2}{(\Sigma X_{ik})^2}$$

For the purpose of these calculations, the catch data were combined for all three sampling events and for all sites within a site type. Indices were calculated to examine similarities between combined catch data from all four repair types (i.e., All Repair = No Bench, Bench, Wetland Trench, 10:1 Slope) and either Naturalized or Unmitigated. Then, to compare species composition between sampling events, Morisita-Horn indices were calculated between events for each of the six site types. An index value of 0 indicates completely distinct samples (with no species in common) and an index of approximately 1 indicates that the species composition of the samples is identical (Horn 1966). The community composition indices are parameters to measure similarity, and should not be interpreted as statistical comparisons (Horn 1966).

5.1.2 Salmon Presence and Abundance (Use) at Sites and Microhabitats within Sites, and Relative Risk of Predation to Juvenile Salmon at Sites (Fish Objective 2)

In order to determine Chinook salmon abundance at repair sites and microhabitats within sites, we examined the association between abundance and a suite of habitat and environmental variables through Generalized Linear Model (GLM) and General Linear Mixed Model (GLMM) analyses. Since Bench and 10:1 design types are similar (all are bench-type designs) and there was only one Bench Site, we combined these types and used the term “10:1 Slope/Bench” in our analysis.

GLMs and GLMMs were used to analyze the catch data from the three electrofishing events in 2012, with the intention to

- (1) determine the difference in Chinook salmon abundance (interpreted as habitat preference) among the different repair types (with repair type as the categorical predictor variable).
- (2) assess whether specific microhabitat features (such as flow, depth, density of emergent vegetation, shade) are associated with Chinook salmon abundance.
- (3) investigate the relative risk of predation to juvenile salmonids (assumed to increase with abundance of predatory species greater than 150 mm TL) at various repair sites and its relationship to various habitat characteristics.
- (4) determine if size distributions of Chinook salmon (FL) found at different site types are related to repair type and microhabitat

The models used in this analysis often included an interaction term between two or more of the explanatory variables, generally denoted by “x” in model descriptions (e.g., flow x depth). In this analysis, both numerical variables (e.g., counts, depths, percent shade...) and categorical variables are used in model fitting. Categorical variables are represented in the model by a dummy variable.

The process of model fitting and selection can be best described as a three-step process:

- During first step of data analysis, models were developed that incorporate the effects of various spatial and temporal components (e.g., river region, time of sampling event) as well as the different design types (described earlier in this report) on the abundance of focal fish species. Habitat Features (such as vegetation, woody material or shade) are not included in this first step to avoid confounding their effects with those of design type, which complicates the interpretation of the model output.
- During the second step of data analysis, a different set of models was developed, incorporating the effects of different Habitat Features in addition to temporal and spatial variables. Repair types were not included in these models to independently interpret habitat effects on fish abundance.

- Effects of both, different repair types and habitat features were included in the third set of models (in addition to the spatial and temporal variables).

Through this process significant differences in fish distribution can be discerned, and relationships between fish abundance and habitat types (or habitat features) can be detected, while controlling for the confounding effects of variables not related thereto (such as sampling events or river region). When investigating the effects of habitat features, a suite of models is fit to the data, including a broad variety of predictor categories and their interactions.

Note that for the purpose of this report, catch data for the three sampling events were combined in all models, but month was retained as a categorical predictor variable in all model fits to account for abundance variation related to the timing of the sampling events. Furthermore, while a variety of piscivorous species were encountered during these surveys, black bass (*Micropterus* spp.), striped bass (*Morone saxatilis*), and Sacramento pikeminnow (*Ptychocheilus grandis*) were combined under a single category in one of the models relating to predator abundance.

Model Summary

Site Type

A GLM model of fry and juvenile abundance (fry < 50mm FL <juvenile, residuals follow a Poisson distribution) was first used to assess whether Site Type affects abundance. Site Types used in this model include Mitigated Repair Sites (categories: 10:1 Slope/Bench, No Bench, Wetland Trench), Unmitigated Repair Sites, and Naturalized Sites. To determine if Site Type has a significant effect on fry abundance after accounting for variability introduced by other spatial and temporal variables, this model also included

- Region (Ko-Ket, Garcia Bend, and Knights Landing)
- Event (sampling was conducted on three occasions)
- Region x Month interaction
- Site Type
- Site Type x Month interaction

Mitigated Repair Site Type

A GLM model of was then used to assess whether abundance of Chinook salmon fry and/or juveniles, and predator abundance is affected by Mitigated Repair Site Type. Mitigated Repair Site Types used in this model include Mitigated Repair Sites (categories: 10:1 Slope, Bench, No Bench, Wetland Trench), and Naturalized Sites. To determine if Mitigated Repair Site Type has a significant effect on fry abundance after accounting for variability introduced by other spatial and temporal variable, this model also included

- Region (Ko-Ket, Garcia Bend, and Knights Landing)
- Event (sampling was conducted during three events between March and April)
- Region x Month interaction

Microhabitat

The GLM models used to evaluate the use of different microhabitats by Chinook salmon and fish predators accounted for the following habitat variables:

- Flow (mean flow in feet per second at the sampling site)
- Depth (mean depth at the sampling site in feet)
- Shade (present/absent)
- Emergent Vegetation (absent, low/medium/high density)
- Emergent Woody Material
- Terrestrial Vegetation (percent coverage)
- Terrestrial Woody Material (percent coverage)

To determine whether microhabitat has a significant effect on fry abundance after accounting for variability introduced by other spatial and temporal variable, this model also included

- Region (Ko-Ket, Garcia Bend, and Knights Landing)
- Event (sampling was conducted during three events between March and April)
- Region x Month interaction

The habitat variables analyzed in this report largely overlap with those included in analyses conducted in previous years (H.T. Harvey & Associates and PRBO 2011), yet differences in habitat measurements only allow for comparison and interpretation of results rather than a combination of data across years and subsequent meta-analysis. For example, analyses of fish abundance in this report include shade as a categorical variable, yet in past years the percentage of shade cover was used in analyses. Furthermore, some habitat features included in past analyses (such as submerged vegetation or substrate composition) were not measured during the 2011 and 2012 surveys. Also, even though point measurements of physical habitat (e.g., IWM diversity and dominant IWM size) are available from past surveys, these measurements do not directly correspond to points sampled in subsequent years. Therefore, inclusion of these habitat measurements in this analysis would render the validity of subsequent inferences questionable.

However, this analysis does incorporate effects of some variables not taken into account previously (e.g., flow and density of emergent woody material). Ultimately, while some differences exist between data collection and analysis in 2011/2012 compared to years past, this report offers valuable insights on the overall effectiveness of various bank repair types and a suite of biologically relevant habitat features, and allows for the interpretation of relevant results in a broader temporal framework.

Evaluation of fish habitat preference based on 2-year cumulative monitoring data

To investigate whether patterns of habitat use observed in individual monitoring years (2011 and 2012) are consistent through time and to allow for increased statistical power (due to higher sample numbers), point specific catch data from both monitoring years were combined for further analysis. Analysis steps are similar to those described in previous sections, with the exception that bank slope and current gradient were included as habitat variables of interest. Mean bank slope was estimated for each site and sampling event by performing a linear

regression analysis of recorded average depth on distance from shore where these measurements were taken (i.e., 5, 10 and 15 / approx. 1.5, 3 and 4.5 m) from the wetted margin, regression forced through the origin). Inclusion of this variable, in addition to mean site depth, is intended to provide additional information on the importance of substrate gradient (e.g., do salmonids prefer near-bank habitat in close proximity to deeper water migration corridors, as indicated by a significantly positive coefficient for bank slope?). Similarly, current gradient was estimated for each site and sampling event to determine if the proximity of slow-water refuges to migration corridors (or, in general, habitat with elevated current speed) is a significant factor in determining habitat suitability of monitored locations.

The general analysis protocol followed the procedure described in the preceding sections, and model summaries and results are presented in Section 6.1.

5.2 Sampling with Alternative Collection Methods

FISHBIO used prepositioned area electrofishing, Breeder traps, and quatrefoil light traps on June 20, 2012. Sampling was conducted upstream of Knights Landing, CA (within or in the vicinity of monitoring sites at RM 90.8R and RM 91.9R) under the following environmental conditions:

- Water temperature: 70.3 F
- Conductivity: 130 μ S/cm
- Dissolved oxygen: 8.75 ppm
- Turbidity: 9.45 NTU
- Secchi depth: 3.0 feet/ 0.91 m (shade)

Low catch rates did not permit quantitative analysis of catch data; however, observations are presented in Section 6.2.

5.3 Habitat Monitoring (Habitat Objective 1)

SAM habitat monitoring efforts were conducted in the summer and fall of 2012 and will be reported separately by the USACE (in preparation).

5.3.1 Photo Documentation

Panoramic images of project sites were taken in 2012 and are presented separately, along with 2011 baseline panoramic photos, in digital format (separate CD).

5.3.2 Sidescan Sonar

During the SSS surveys, individual screen shots were saved with a unique file name corresponding to a waypoint in order to capture a series of consecutive overlapping images. At the end of the survey, all screen shots and associated waypoint data was transferred to an SD

card that was later used to transfer data to a desktop computer where the resulting images were processed (cropped) in batches using Infranview graphic software (<http://www.irfanview.com>). Next, software tools (developed by Kaeser and Litts (2010) operating within ArcGIS) were used to identify overlap points in consecutive images by matching rows of pixels with similar values within a user-defined tolerance. The images were then georeferenced using a spline generated from the boat track recorded with the Trimble GPS. The resulting individual georeferenced images were then combined into a single sonar image mosaic (SIM), comprised of pixels with an approximate 10 cm resolution. According to Kaeser and Litts (2010), interpretation of sonar imagery has the potential to allow identification of IWM (≥ 10 cm diameter and ≥ 1.5 m in length), substrate, and bank boundaries. Furthermore, using measuring tools available in ArcGIS, measurements of IWM jam area and count can be produced from quality images with the potential to measure individual IWM pieces length. Positional accuracy of methods described in Kaeser and Litts (2010) was ± 5.95 m.

Electrofishing activities were recorded by a video camera mounted to the front rail of the boat with the general fish capture area oriented within the field of view. A GPS position of the boat was displayed on the video using an overlay interfaced with the GPS/chartplotter (Humminbird, Eufaula, AL; 1197c). Personnel netting fish called out species, life stage, and count of fish captured. Videos were later reviewed and catch information and associated GPS position were input to an Excel spreadsheet and imported to GIS where the results were overlaid with the SSS imagery with the intention of documenting fish association with habitat features.

6.0 RESULTS

6.1 Fisheries Monitoring (Fish Objectives 1 and 2)

Three sampling events were conducted in 2012: March 17-20, April 2-4 and April 16-18. The data obtained from electrofishing efforts at each sampling site included environmental conditions (Appendix A) and species counts (Appendix B).

Fish species presence (Fish Objective 1) and fish community abundance and composition (Fish Objective 2) are presented in section 6.1.1. Models used to evaluate habitat preference of Chinook salmon and piscivorous predators and subsequent relevant interpretations (Fish Objective 2) are presented in Section 6.1.2.

6.1.1 Species Captured (Fish Objective 1) and Fish Community Abundance and Composition (Fish Objective 2)

A total of 29 fish species were collected in the three sampling events (Table 2). Overall, focus fish species recorded (captured and missed) included 1,362 juvenile Chinook salmon (849 captured and 513 missed), 24 juvenile steelhead (19 captured, 5 missed), and 49 delta smelt (49 captured). Predatory species included 215 black bass (177 captured, 38 missed), 12 striped bass (nine captured, three missed), 153 Sacramento pikeminnow (140 captured, 13 missed), and 120 inland silverside (120 captured)³. Species counts for each sampling site and event are provided in Appendix B. Brief summaries for fish species categories (focus, predator, and other species) are presented below.

Focus Fish Species by Sampling Event

First sampling event

During the first sampling event (March 17-20), a total of 289 juvenile (18 fry and 271 parr/smolt) Chinook salmon were captured. Assignment of measured Chinook salmon to race (using Fisher race tables) indicated that 31% (n= 90) were spring-run and 69% (n= 198) were fall-run fish, and less than 1% (n=1) were winter-run. Approximately 6.2% (n=18) of measured Chinook salmon were fry, 51.2% (n=148) were classified as parr, and 42.6% as smolt (n=123). Mortality rates were low during this survey, with an overall rate of 3.8% (11 individuals) for juvenile Chinook salmon (3.3% and 4.0% for spring- and fall-run, respectively). Detailed length-frequency distributions are depicted in Figures 9 and 10.

In addition, 14 juvenile steelhead were captured with an average length of 206.6 mm FL (range: 127-262 mm). Eleven of the steelhead were of hatchery origin. No mortalities occurred.

Seventeen delta smelt were captured during the March sampling event, with a mean length of 69.2 mm FL (range: 65-75 mm). No mortalities occurred.

³ Inland silversides are potential predators of delta smelt eggs and/or larvae. However, sampling was not conducted in delta smelt spawning habitat downstream of RM 20, so these fish are unlikely to function as predators in the sampling reach. Nonetheless, data was provided regarding their presence in sampled sites.

Second sampling event

During the second sampling event (April 2- 4), a total of 356 juvenile (83 fry and 273 parr/smolt) Chinook salmon were captured. Based on the Fisher race tables, 253 of measured Chinook salmon were fall-run, 98 were spring-run, and five were winter-run Chinook salmon. Approximately 23.3% of the Chinook salmon measured during this event were classified as fry (n=83), 54.2% (n=193) classified as smolts, and 22.5% (n=80) as parr. There were 21 Chinook salmon mortalities (with an overall mortality rate of 5.9%), consisting of 14 fall-run fish (5.5% of fall-run catch) and seven spring-run Chinook salmon (7.1% of spring-run catch). Detailed length-frequency distributions are depicted in Figures 9 and 10.

In addition, three steelhead were captured during this sampling event (one of them of hatchery origin), with an average length of 217.3 mm FL. No mortalities occurred.

Twenty delta smelt were captured during the second sampling event, with a mean length of 68.1 mm FL (range: 63-73 mm FL). One mortality occurred.

Third sampling event

During the third sampling event (April 16-18), a total of 204 juvenile Chinook salmon were captured (28 fry and 176 parr/smolt). Of these fish, 172 were considered fall-run, 31 spring-run and one individual was classified as late fall-run Chinook salmon. Approximately 4.4% of the Chinook salmon measured during this event were classified as fry (n=9), 26.5% (n=54) classified as smolts, and 69.1% (n=141) as parr. There were 18 Chinook salmon mortalities (overall mortality rate of 8.8%), consisting of 15 fall-run fish (8.7% of fall-run catch), two spring-run fish (6.5% of spring-run catch) and one late fall-run individual.

Two juvenile steelhead were captured during this event (one of them of hatchery origin), with a mean length of 218 mm FL. No mortalities occurred.

Twelve delta smelt were captured during the third sampling event, with a mean length of 70.2 mm FL (range: 65-75 mm). One mortality occurred.

Table 2. Summary of all species captured during 2009-2012.

Common (Scientific) Names	Native Status (N/ I) ¹	Counts				Mean Total Lengths ²			
		2009	2010	2011	2012	2009	2010	2011	2012
Focus Species									
Chinook salmon (<i>Oncorhynchus tshawytscha</i>)	N	158	464	405	849	64	61	71	69
Steelhead/rainbow (<i>Oncorhynchus mykiss</i>)	N	6	9	7	19	200	227	175	220
Delta smelt (<i>Hypomesus transpacificus</i>)	N	0	2	6	49	-	76	80	74
Predatory Species									
Sac. pikeminnow (<i>Ptychocheilus grandis</i>)	N	25	52	79	140	155	177	142	153
Striped bass (<i>Morone saxatilis</i>)	I	4	10	8	9	260	236	423	448
Black bass-Large and smallmouth (<i>Micropterus salmoides/ dolomieu</i>)	I	99	137	131	177	211.5	145.5	143	169
Inland silverside (<i>Menidia beryllina</i>)	I	96	160	63	120	74	61	63	68
Other species									
Pacific lamprey (<i>Lampetra tridentata</i>)	N	1336	811	1	-	n/a	n/a	-	-
Threadfin shad (<i>Dorosoma petenense</i>)	I	1	1	4	8	103	84	101	104
American shad (<i>Alosa sapidissima</i>)	I	-	-	-	2	-	-	-	109
Hitch (<i>Lavinia exilicauda</i>)	N	3	5	3	13	274	250	100	83
California roach (<i>Lavinia symmetricus</i>)	N	-	-	2	-	-	-	58	-
Sac. blackfish (<i>Orthodon microlepidotus</i>)	N	-	15	3	1	-	426	425	470
Sac. splittail (<i>Pogonichthys macrolepidotus</i>)	N	1	22	5	72	290	341	369	143
Hardhead (<i>Mylopharodon conocephalus</i>)	N	-	-	-	1	-	-	-	408
Golden shiner (<i>Notemigonus crysoleucas</i>)	I	5	13	3	11	77	52	88	82
Red shiner (<i>Cyprinella lutrensis</i>)	I	-	-	1	1	-	-	54	55
Goldfish (<i>Carassius auratus</i>)	I	9	5	-	2	233	176	-	208
Common carp (<i>Cyprinus carpio</i>)	I	11	12	12	24	485	488	390	560
Sacramento sucker (<i>Catostomus occidentalis</i>)	N	33	11	43	29	170	266	147	358
Black bullhead (<i>Ameiurus melas</i>)	I	-	1	-	1	-	169	-	203
Brown bullhead (<i>Ameiurus nebulosus</i>)	I	5	-	-	-	83	-	-	-
Yellow bullhead (<i>Ameiurus natalis</i>)	I	2	-	-	-	58	-	-	-
White catfish (<i>Ameiurus catus</i>)	I	10	38	2	-	195	224	197	-
Channel catfish (<i>Ictalurus punctatus</i>)	I	2	9	-	-	200	306	-	-
Wakasagi (<i>Hypomesus nipponensis</i>)	I	-	-	-	6	-	-	-	81
Western mosquitofish (<i>Gambusia affinis</i>)	I	2	3	-	1	34	30	-	35
Prickly sculpin (<i>Cottus asper</i>)	N	-	-	11	35	-	-	92	88
Bluegill sunfish (<i>Lepomis macrochirus</i>)	I	14	75	14	37	92	70	86	84
Redear sunfish (<i>Lepomis microlophus</i>)	I	-	15	9	4	-	159	117	181
Pumpkinseed sunfish (<i>Lepomis gibbosus</i>)	I	16	-	-	-	121	-	-	-
Green sunfish (<i>Lepomis cyanellus</i>)	I	5	19	3	7	106	92	121	88
Warmouth (<i>Lepomis gulosus</i>)	I	1	14	3	1	118	63	77	122
Black crappie (<i>Pomoxis nigromaculatus</i>)	I	1	1	-	3	174	166	-	113
Spotted bass (<i>Micropterus punctulatus</i>)	I	-	1	-	-	-	245	-	-
Bigscale logperch (<i>Percina macrolepida</i>)	I	1	-	-	-	85	-	-	-
Tule perch (<i>Hysterocarpus traski</i>)	N	310	376	38	113	113	112	99	115

¹ N = native, I = introduced; ² Lengths are provided as fork lengths for all species with forked or truncated caudal fins, otherwise total length was used

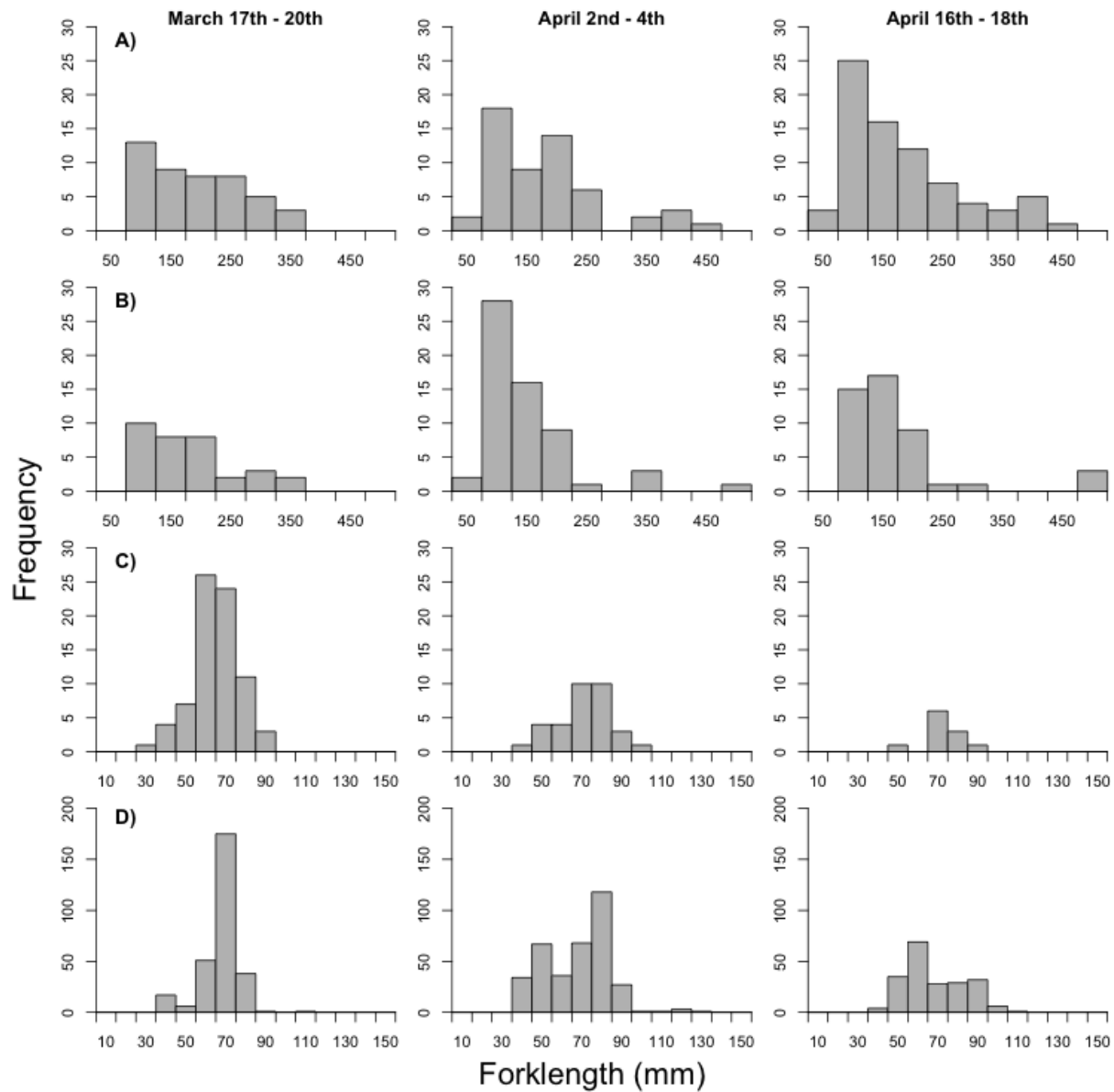


Figure 9. Comparison of length-frequency histograms between 2012 sampling events (by column) for A) black bass, B) Sacramento pikeminnow, C) inland silversides and D) Chinook salmon.

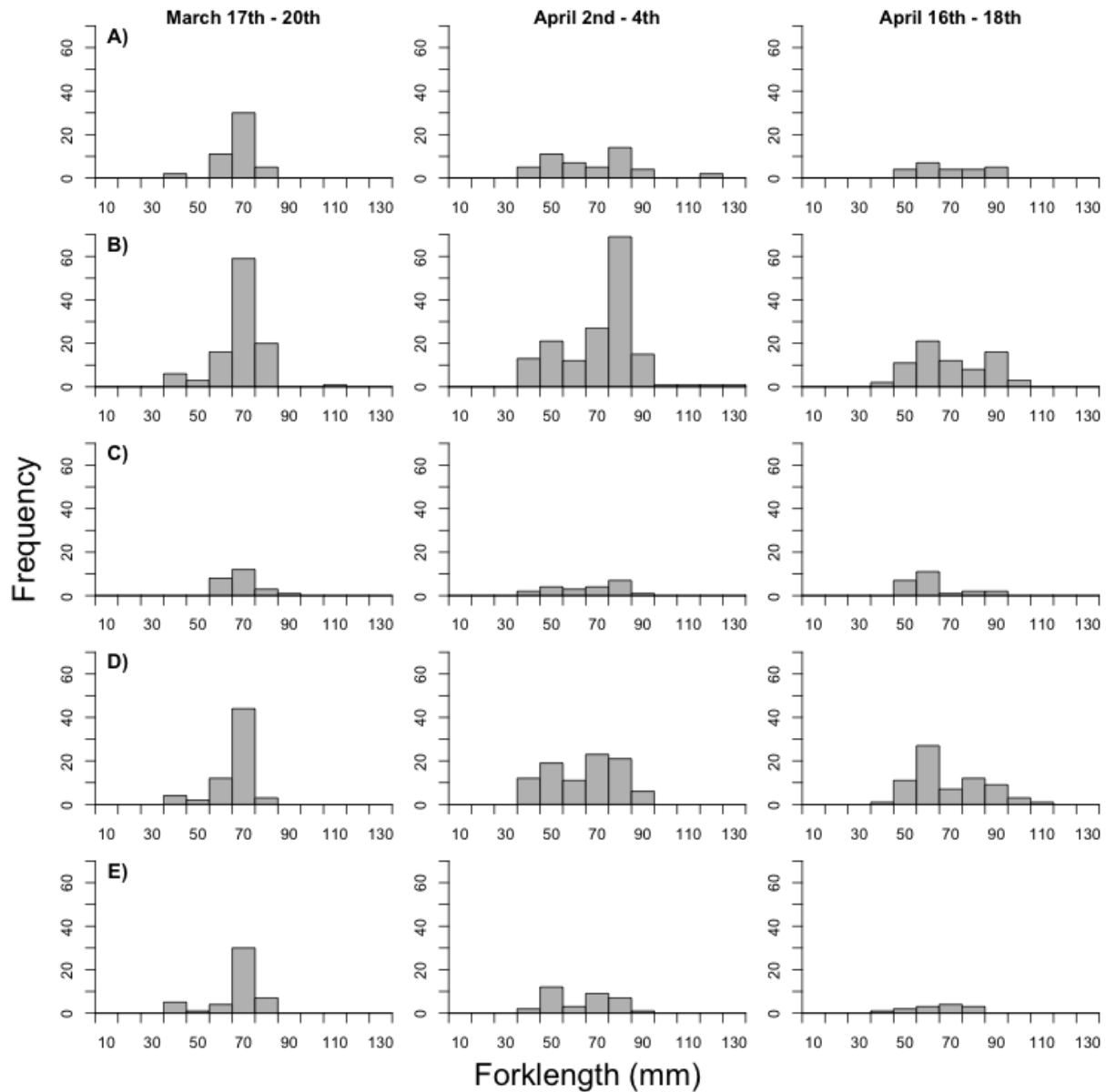


Figure 10. A comparison of length-frequency histograms between 2012 sampling events (by column) for Chinook salmon at different site types: A) Unmitigated Repair, B) Naturalized, C) No Bench, D) 10:1 Slope/Bench, and E) Wetland Trench.

Predatory Fish Species. During the first sampling event, a total of 158 predatory fish were captured: 46 black bass, 33 Sacramento pikeminnow, 76 inland silverside, and three striped bass.

During the second sampling event, a total of 151 predatory fish were captured: 55 black bass, 61 Sacramento pikeminnow, 33 inland silverside, and two striped bass.

During the third sampling event, a total of 137 predatory fish were captured: 76 black bass, 46 Sacramento pikeminnow, 11 inland silverside, and four striped bass.

Length frequency distributions by site type and sampling event for bass predators and combined piscivores are depicted in Figures 11-12.

Other Fish Species. During the three sampling events, a total of 121, 128, and 125 fish belonging to other species were captured, respectively. During the first event catch of non-focal species consisted to 79% (95/121) of native fishes, and was numerically dominated by Sacramento splittail (34% or 41/121) and Tule Perch (28% or 34/121). Incidental (non-focal) fish catches during the second event consisted to 63% (80/128) of native fishes, with tule perch (39%, or 40/128) and Sacramento splittail (16%, or 21/128) comprising the bulk of the incidental catch. During the third sampling event, catch of non-focal species consisted to 74% of native fishes (92/125), with tule perch (24%, or 30/125) and prickly sculpin (22%, or 27/125) being numerically predominant. A detailed catch summary of non-focal species can be found in Appendix B.

Morisita-Horn Indices

Morisita-Horn indices for pair-wise site type comparisons indicate that species composition was very similar between most Site Types: No Bench, 10:1 Slope, Bench, and Wetland Trench (Table 3)⁴ and Repair Types (Table 4). In general, index values were high, indicating a large degree of overlap or similarity between Site Types. The largest dissimilarities (though still relatively minor) were observed between Naturalized and No Bench Sites, as well as between Naturalized and Wetland Trench Sites. This appears to be largely due to increased predator abundance at the No Bench and Wetland Trench Sites.

A comparison of species composition between sampling events indicates that species composition within the No Bench, Wetland Trench, and Unmitigated Repair Sites were least consistent through time, with the largest differences in fish communities observed between the first and third sampling event. In contrast, the community composition of the Naturalized and 10:1 Slope/Bench sites was characterized by temporal stability (Table 5).

⁴ NOTE: The community composition indices are parameters of similarity and indicate the degree of overlap between two samples; the parameters should not be interpreted as statistical comparisons (Horn 1966).

Table 3. The Morisita-Horn index of similarity of species composition between pairs of combined site types.

	All Mitigated Repairs	Unmitigated Repairs
Unmitigated Repairs	0.96	-
Naturalized	0.90	0.95

Note: The index is 0 when the samples have no species in common and 1 when the samples have the same proportional species composition

Table 4. The Morisita-Horn index of similarity of species composition between pairs of site types.

	10:1 Slope/Bench	Unmitigated Repair	Naturalized	No Bench
Unmitigated Repair	0.94	-	-	-
Naturalized	0.90	0.95	-	-
No Bench	0.96	0.95	0.88	-
Wetland Trench	0.94	0.94	0.87	0.96

Note: The index is 0 when the samples have no species in common and 1 when the samples have the same proportional species composition

Table 5. Morisita-Horn index of similarity of species composition between sampling events at each site type.

Event	Unmitigated Repair	Naturalized	No Bench	10:1 Slope/Bench	Wetland Trench
1 & 2	0.99	0.95	0.82	0.92	0.92
1 & 3	0.86	0.93	0.78	0.90	0.68
2 & 3	0.90	0.94	0.96	0.96	0.84

Note: The index is 0 when the samples have no species in common and 1 when the samples have the same proportional species composition

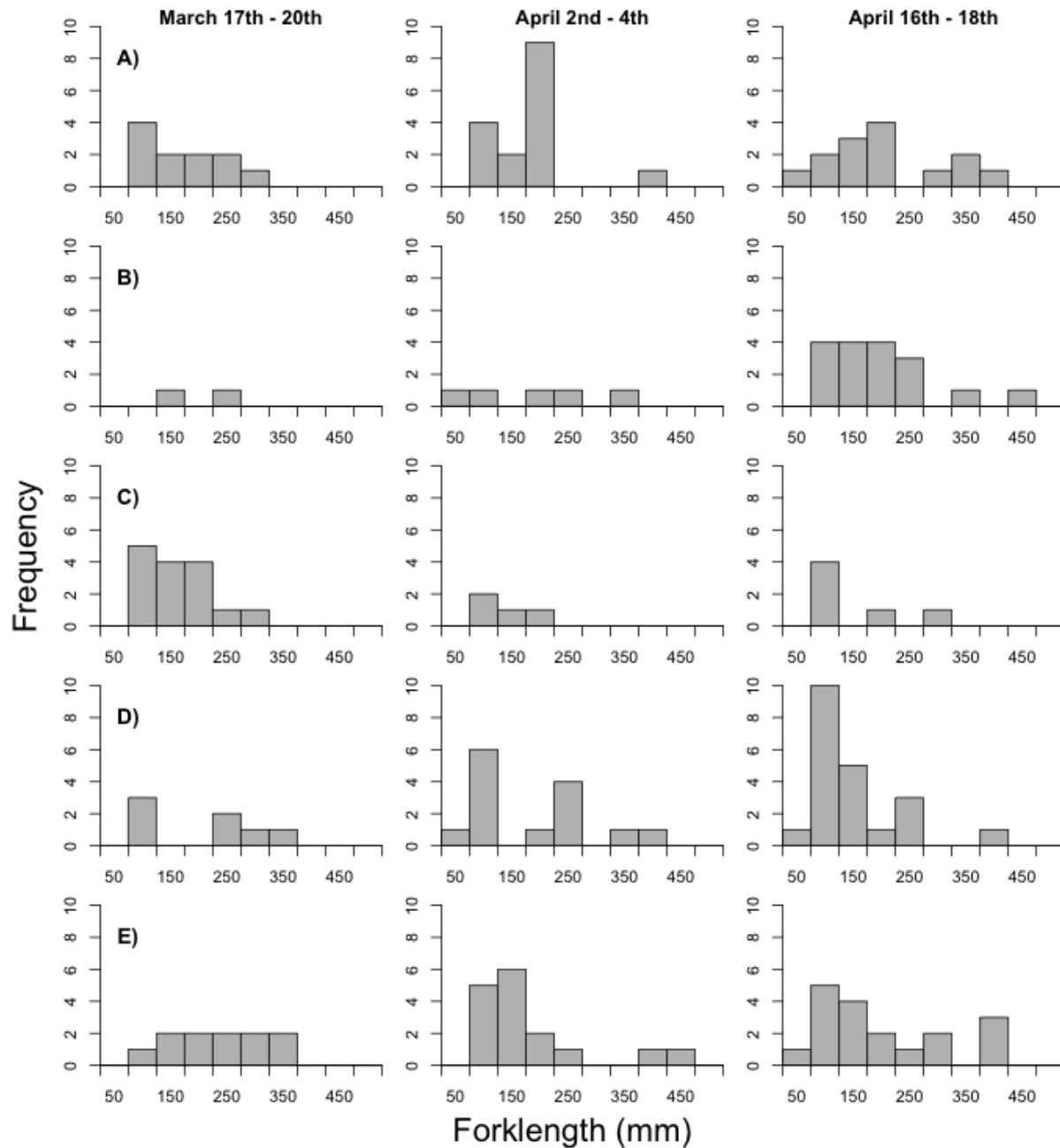


Figure 11. A comparison of length-frequency histograms between 2012 sampling events for Black Bass at A) Unmitigated Repair, B) Naturalized, C) No Bench, D) 10:1 Slope/Bench, and E) Wetland Trench sites.

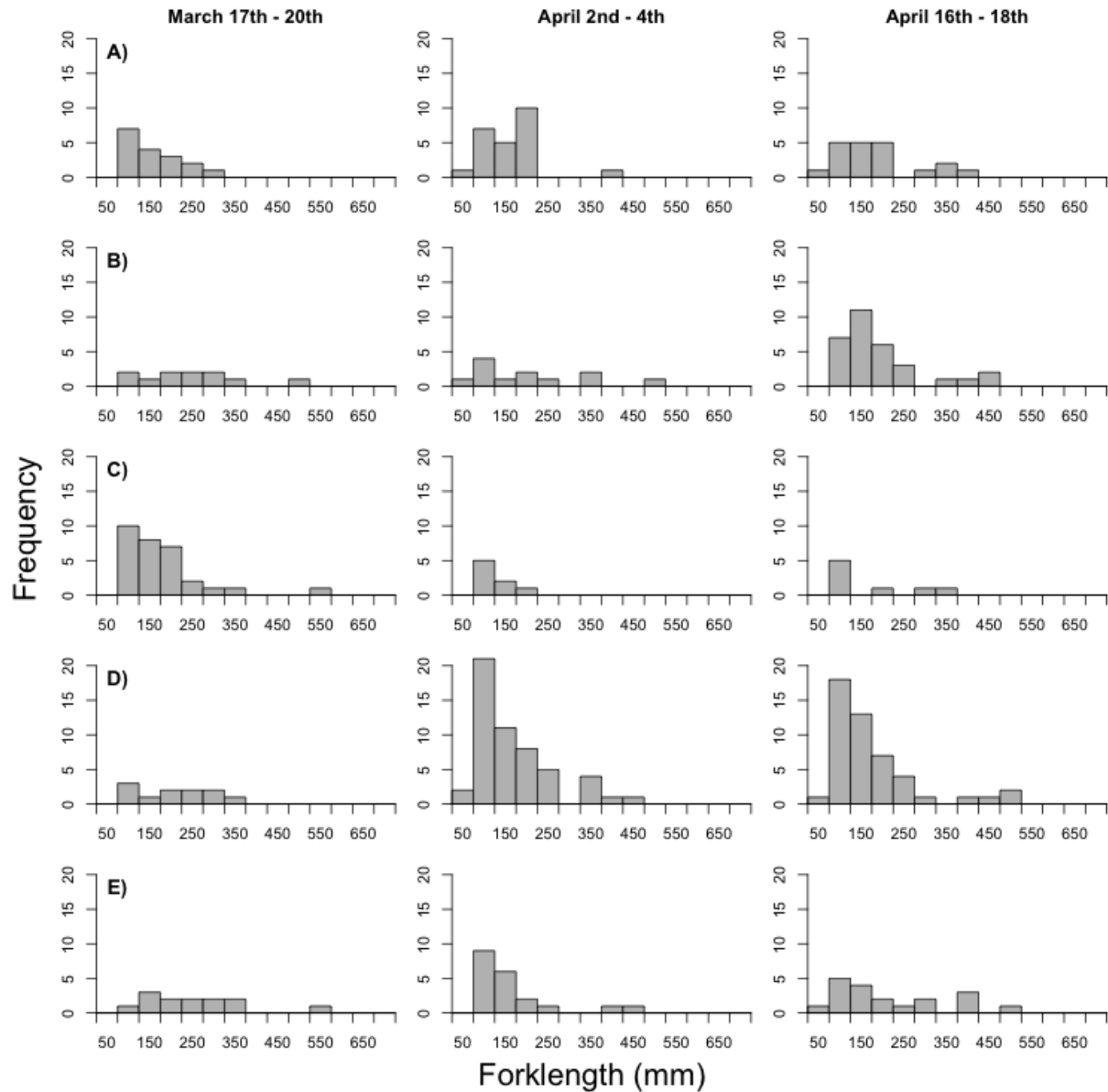


Figure 12. A comparison of length-frequency histograms between 2012 sampling events for piscivorous species (black bass, striped bass and Sacramento pikeminnow) at A) Unmitigated Repair, B) Naturalized, C) No Bench, D) 10:1 Slope/Bench, and E) Wetland Trench sites.

6.1.2 Salmonid Presence, Abundance (Use) and Relative Risk of Predation to Juvenile Salmonids at Monitoring Locations and Associated Microhabitats (Fish Objective 2)

Count data used in these analyses contains 1,165 sampling points, representing samples for 21 sampling sites and six different site types. A summary of results relevant to the evaluation of habitat use by Chinook salmon and predator fish is presented below. An underlying assumption for interpretation of these results is that fish abundance is reflective of habitat preference.

While a number of biologically important measurements were evaluated in these analyses, a multitude of additional factors may influence fish abundance and habitat preference. In the subsequent interpretation, significant effects having negative coefficients are considered to indicate lower fish abundance than the Reference Category, while those having positive coefficients indicate higher fish abundance than the Reference Category and are indicative of habitat preferences.

Site Types

In the first step, the model incorporated the effect of Site Type (Mitigated Repair, Unmitigated Repair, and Naturalized) to determine whether different types of repairs (i.e., Unmitigated versus Mitigated Repairs) influence Chinook salmon fry and juvenile abundance in comparison to each other and with Naturalized Sites.

The predictive equations for evaluating the effect of Site Type on Chinook salmon fry and juvenile abundance were in the general form

$$\log(c) = \beta_0 + \beta_{1:2}RegionA:B + \beta_{3:4}Event_{Mid:Late} + \beta_{5:6}SiteA:B + \beta_{7:10}Region \times Event$$

where c is predicted Chinook salmon abundance (fry or juvenile), Regions A and B correspond to River Regions Garcia Bend (RM40-70) and Knights Landing (RM70-95). Ko-Ket (RM21-39), the lowermost river region, was used as the reference region. *Event* represents the sampling time (the first sampling event “*early*” was used as the reference point, *mid* and *late*), Sites A and B correspond to the Site Types not used as the reference category in each respective model, and *Region x Month* refers to the interaction between region and sampling time (four combinations, using the first sampling event as the reference point).

Coefficients for the different Sites Types (regardless of the reference category used in the model) indicate that abundance of Chinook salmon fry and juveniles was highest at Naturalized Sites. While there was no significant difference in fry abundance between Unmitigated and Mitigated Repair Sites, the inability to discern habitat preferences may be due to the relatively low sample size of Chinook salmon fry (only 103 fry were captured in all three sampling events). Juvenile abundance, however, was significantly higher at Mitigated Repair Sites than at Unmitigated Repair Sites (Table 6). Therefore, data from Unmitigated Repair Sites was excluded from further analysis to focus on the parameters of main interest in this study.

Table 6. Influence of combined Site Type on abundance of Chinook salmon (sampled in March and April 2012).

Reference Category	Site Type	<u>Fry</u>			<u>Juvenile</u>		
		Coefficient	SE	Sig.	Coefficient	SE	Sig. ¹
Unmitigated	Mitigated	0.39	0.30	NS	0.58	0.09	***
	Naturalized	0.67	0.31	*	0.88	0.09	***
Mitigated Repair	Unmitigated	-0.39	0.30	NS	-0.58	0.09	***
	Naturalized	0.28	0.23	NS	0.30	0.06	***
Naturalized	Unmitigated	-0.67	0.31	*	-0.88	0.09	***
	Mitigated	-0.28	0.23	NS	-0.30	0.06	***

¹Significance categories: “NS” (not significant), “*” (p<0.05), “**” (p<0.01), and “***” (p<0.001)

Mitigated Repair Site Types

General linear mixed models (GLMMs) were used to assess whether individual Mitigated Repair Site Types (10:1 Slope/Bench, No Bench, Wetland Trench) influence Chinook salmon fry and/or juvenile abundance. General linear mixed models are used to account for the non-independence between observations (i.e., multiple points were sampled within each site, and the resulting catches are likely correlated). Naturalized Sites were used as the reference category in all model fits.

The predictive equations for evaluating the effect of Mitigated Repair Site Types on Chinook salmon abundance were in the general form

$$\log(c) = \beta_0 + \beta_{1:2}RegionA:B + \beta_{3:4} + \beta_{5:7}TypeA:C + \beta_{8:11}Region|Event$$

where c is predicted Chinook salmon abundance (fry or juvenile), Regions A and B correspond to regions Garcia Bend (RM40-70) and Knights Landing (RM70-95), respectively (the reference region for this model was Ko-Ket [RM21-39]), $Event$ represents the sampling time, Types A through C correspond to repair types 10:1 Slope/Bench, No Bench, and Wetland Trench, respectively (Naturalized Sites were used as the reference category in all model fits), and $Region \times Event$ refers to the interaction between region and sampling month. Catch of Chinook salmon fry and juveniles (all three sampling events combined) is illustrated in Figures 13 and 14, respectively.

Fry

In this model, no significant abundance difference between Naturalized Sites and the 10:1 Slope/Bench or Wetland Trench Sites could be discerned; however, Chinook salmon fry abundance was significantly lower at No Bench Sites when compared to Naturalized Sites (Table 7).

Juveniles

Model output predicts highest abundance of juvenile Chinook salmon at 10:1 Slope/Bench Sites, followed by Naturalized Sites. Similar to 2011 monitoring results, Wetland Trench Sites also had a significant effect in this model, and were associated with reduced abundance for juvenile Chinook salmon (Table 7). No significant differences in abundance could be detected between Naturalized and No Bench Sites in 2011 or 2012.

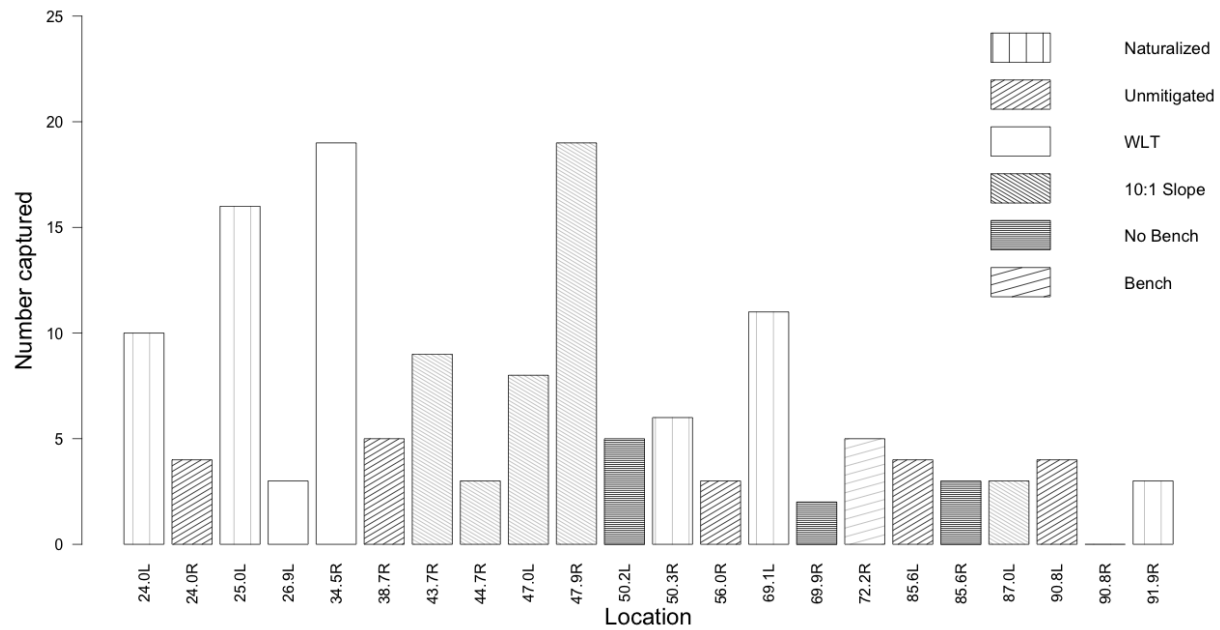


Figure 13. Chinook salmon fry captured at each Site Type in March and April 2012

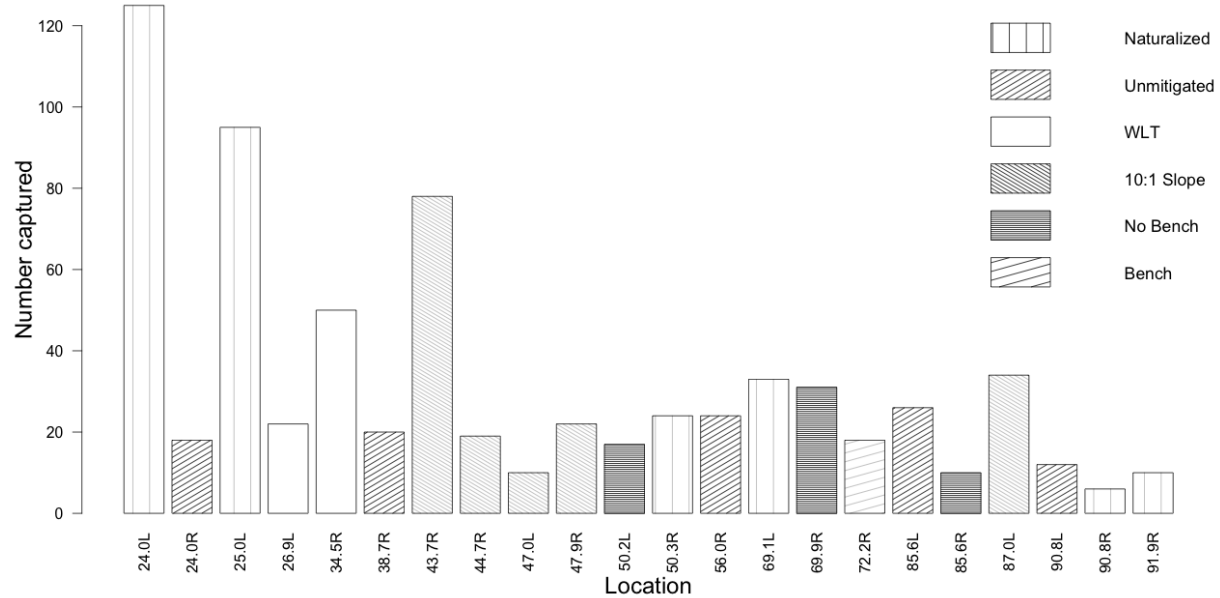


Figure 14. Chinook salmon juveniles captured at each Site Type in March and April 2012.

Table 7. Influence of Mitigated Repair Site Type (on abundance of Chinook salmon fry and juveniles in March and April 2012.

Reference Category	Site Type	<u>Fry</u>			<u>Juvenile</u>		
		Coefficient	SE	Sig.	Coefficient	SE	Sig. ¹
Naturalized	10:1 Slope/Bench	-0.57	0.21	NS	0.55	0.11	***
	No Bench	-1.16	0.47	*	0.03	0.14	NS
	WLT	0.21	0.33	NS	-0.93	0.11	***

¹Significance categories: “NS” (not significant), “*” (p<0.05), “**” (p<0.01), and “***” (p<0.001)

Habitat Features

To investigate potential relationships between Chinook salmon fry and juvenile abundance and various microhabitat features, the model used in this step included the following habitat variables (in addition to the spatial and temporal variables included in all models): shade (present/absent), density of emergent vegetation (absent/low/medium/high, absent was used as the reference category), density of emergent woody material (absent/low/medium/high, absent was used as the reference category), flow (feet/s), depth (feet), terrestrial vegetation (TV, percent bank coverage) and terrestrial woody material (percent bank coverage).

The predictive equations for evaluating the effect of habitat features on Chinook salmon abundance were in the general form

$$\begin{aligned} \log(c) = & \beta_0 + \beta_{1:2}RegionA:B + \beta_{3:4}Event_{Mid:Late} + \beta_5Shade_{present} + \dots \\ & + \beta_{6:8}EV_{dens.} + \beta_{9:11}EW_{dens.} + \beta_{12}Flow + \beta_{13}Depth + \beta_{14}TV\% + \dots \\ & \beta_{15}TW\% + \beta_{16:19}Region|Event \end{aligned}$$

where c is predicted Chinook salmon abundance (fry or juvenile), Regions A and B correspond to regions Garcia Bend (RM40-70) and Knights Landing (RM70-95), respectively (the reference region for this model was Ko-Ket [RM21-39]), $Event$ represents the sampling time, $Shade$ refers to shade presence, EV and EW correspond to the density of emergent vegetation and emergent woody material, respectively (low, medium and high, absence is used as the reference category), TV and TW refer to the percent coverage of terrestrial vegetation and terrestrial woody material, respectively, and $Region|Month$ refers to the interaction between region and sampling month (four combinations, excluding the reference region).

Fry

Of the habitat variables analyzed in this model, only average depth and flow appeared to have a significant effect on the abundance of Chinook salmon fry (Table 8).

Fry abundance was negatively correlated to flow, indicating higher Chinook salmon fry abundance in areas with slow currents. Interestingly, depth was positively associated with fry

abundance, which indicates that more fry were encountered at sites with deep water than at those with a lesser average depth. However, this result should be considered as inconclusive; results from 2011 monitoring indicated a highly significant negative correlation between fry abundance and depth, an inconsistency that may be attributed to the low overall sample size of fry in 2012 or other, over-riding factors such as flow.

None of the other microhabitat variables (shade, density of emergent vegetation, density of woody material, percent bank coverage of terrestrial vegetation and percent bank coverage of terrestrial woody material) had a significant effect on fry abundance in this model.

Juveniles

Both flow and depth were significantly associated with the abundance of juvenile Chinook salmon (Table 8). The negative coefficients indicate higher juvenile Chinook salmon abundance in shallow habitats with slow currents. In contrast, an inverse association between juvenile abundance and increased current velocity was found in 2011, when juvenile Chinook salmon abundance was higher in areas with faster current velocities. The presence of shade also had a negative effect on the abundance of juvenile Chinook salmon, a factor that did not appear to have a significant effect on habitat use in 2011. Percent bank coverage of terrestrial vegetation and terrestrial woody material at the sampling site appeared to have a significant positive effect on the abundance of Chinook salmon juveniles (Table 8). Though these terrestrial habitat features appear to influence Chinook salmon abundance, the mechanisms of this interaction are unclear. It is possible that these features affect variables not accounted for in the models presented here (such as prey supplementation from adjacent vegetation), or are reflective of associated habitat conditions below the water surface level (diverse habitat on shore continues under water). The latter explanation is supported by a positive association between low and medium levels of emergent vegetation, and juvenile Chinook salmon abundance. Emergent woody material, however, did not significantly influence juvenile abundance (Table 8).

Table 8. Influence of microhabitat features on abundance of Chinook salmon fry and juvenile sampled in March and April 2012.

Habitat Variable	Levels	<u>Fry</u>			<u>Juvenile</u>		
		Coefficient	SE	Sig.	Coefficient	SE	Sig. ¹
Shade	-	0.24	0.24	NS	-0.23	0.07	***
Flow	-	-2.23	0.83	**	-1.05	0.25	***
Depth	-	0.31	0.11	**	-0.11	0.03	***
Terr. Veg.	-	-0.26	0.95	NS	1.36	0.39	***
Terr. WM	-	-0.46	1.07	NS	1.37	0.41	***
Emergent Veg.	Low	0.42	0.37	NS	0.44	0.10	***
	Medium	0.70	0.40	NS	0.35	0.12	**
	High	-0.60	0.56	NS	0.21	0.13	NS
Emergent WM	Low	0.25	0.29	NS	-0.10	0.08	NS
	Medium	-0.30	0.49	NS	-0.05	0.12	NS
	High	-0.82	0.83	NS	-0.26	0.19	NS

¹Significance categories: "NS" (not significant), "*" (p<0.05), "***" (p<0.01), and "****" (p<0.001)

Combined Mitigated Repair Site Types/Habitat Features

To investigate how Chinook salmon fry and/or juvenile abundance is affected by both Mitigated Repair Site Types and habitat features, models that combined these variables were used in this step. Variables included the different Mitigated Repair Site Types (10:1 Slope/Bench, No Bench, Wetland Trench) with Naturalized Sites used as the reference category and those habitat features that were previously determined to have a significant effect on Chinook salmon abundance (Table 9), in addition to the spatial and temporal variables included in all models.

The predictive models for evaluating the effect of Mitigated Repair Site Types and habitat features on Chinook salmon abundance were in the general form

$$\log(c) = \beta_0 + \beta_{1:2}RegionA:B + \beta_{3:4}Event_{Mid:Late} + \beta_{5:7}TypeA:C + \dots$$

$$\beta_8Shade_{present} + \beta_{9:11}EV_{dens.} + \beta_{12:14}EW_{dens.} + \beta_{15}Flow + \beta_{16}Depth + \beta_{17}TV\% + \dots$$

$$\beta_{18}TW\% + \beta_{19:22}Region|Month$$

where c is predicted Chinook salmon abundance (fry or juvenile), Regions A and B correspond to regions Garcia Bend (RM40-70) and Knights Landing (RM70-95), respectively (the reference region for this model was Ko-Ket [RM21-39]); $Month$ represents the sampling time; Types A through C correspond to repair sites 10:1 Slope/Bench, No Bench, and Wetland Trench, respectively; EW correspond to the density of emergent woody material (low, medium and high, absence was used as the reference category); TV and TW refer to the percent coverage of terrestrial vegetation and terrestrial woody material, respectively; and $Region|Month$ refers to the interaction between region and sampling month (four combinations, excluding the reference region). Habitat variables without significant effects in the previous model cannot be significant in this combined model and were therefore not included.

Fry

The combined model used to evaluate Chinook salmon fry habitat associations included design type, flow and depth, in addition to Region and Event (which are included in all models).

Habitat variables with significant effects on fry abundance in the previous modeling step (i.e., flow and depth) also showed a negative correlation in this analysis (Table 9). This indicates that those habitat variables are important in determining fish use despite differences in repair design. However, as discussed earlier, this result should be considered as inconclusive; results from 2011 monitoring indicated a highly significant negative correlation between fry abundance and depth, an inconsistency that can likely be attributed to the low overall sample size of fry in 2012.

Juveniles

The 10:1 Slope/Bench Repair Site Type shows a significant positive effect on juvenile abundance in both the Mitigated Repair Site Type model (Table 7) and this combined Mitigated Repair Site Types/Habitat Features model (Table 9), indicating a strong effect of design type on habitat use of juvenile Chinook salmon. Similarly, Wetland Trench Site Types were associated with reduced Chinook salmon abundance in both models, indicating that variation introduced by environmental factors is minimal compared to the overall effects of this repair site type.

Likewise, the effect of low and medium density Emergent Vegetation was significant in the model evaluating habitat characteristics, as well as in the combined model. Furthermore, terrestrial woody material and vegetation also had consistent positive correlations with juvenile Chinook salmon abundance, implying that intermediate levels of emergent vegetation improve habitat for juvenile salmon and should be considered in the planning effort for future site repairs (no significant relationship was found for high density emergent vegetation). Likewise, the combined model confirmed prediction of higher juvenile abundance in slow current, yet water depth is no longer considered significant in this model. This suggests that slow velocity habitat is an important factor determining Chinook salmon habitat use irrespective of repair design, yet the effects of water depth are confounded by design type, of which depth (bank slope) is an important component.

Table 9. Influence of Repair Site and microhabitat variables on abundance of Chinook salmon sampled in March and April 2012.

Habitat Variable	Levels	<u>Fry</u>			<u>Juvenile</u>		
		Coefficient	SE	Sig.	Coefficient	SE	Sig. ¹
Mitigated Repair Site	10:1 Slope/ Bench	-0.10	0.41	NS	0.61	0.16	**
	No Bench	-0.84	0.51	NS	0.02	0.18	NS
	WLT	-0.01	0.40	NS	-0.78	0.12	***
Flow	-	-1.87	0.84	*	-1.29	0.27	***
Depth	-	0.28	0.14	*	0.05	0.05	NS
Emergent Veg. ²	Low	-	-	-	0.46	0.10	***
	Medium	-	-	-	0.35	0.11	**
	High	-	-	-	0.09	0.12	NS
Terr. Veg. ¹	-	-	-	-	1.23	0.38	**
Terr. WM ¹	-	-	-	-	0.91	0.40	*

¹ Significance categories: “NS” (not significant), “*” (p<0.05), “**” (p<0.01), and “***” (p<0.001)

² Variables identified by superscript are not included in the combined model for fry abundance due to lack of significance in previous analysis steps

Predators

Note that in the following discussion of predator habitat associations, estimates for coefficients for the variables “river region” and “sampling event” are included. While these variables were included in all model fits pertaining to Chinook salmon abundance to account for spatial and temporal variation that cannot be attributed to repair design or microhabitat features, they were not specifically reported or discussed because Chinook salmon abundance is expected to vary due to active downstream migration. For predators, however, coefficient estimates for these variables are reported as they may inform spatial distribution tendencies of these species.

Mitigated Repair Site Type

A GLM was used to evaluate the effects of different Mitigated Repair Site Types on predator abundance. One model focused exclusively on the abundance of black bass, while another was used to evaluate the combined abundance of all potential salmonid piscivores, including black bass, striped bass and Sacramento pikeminnow. Mitigated Repair Site Types evaluated in this model include 10:1 Slope/Bench, No Bench, and Wetland Trench. Naturalized Sites were used as

the reference category in all model fits. Spatial and temporal variables (Region, Month) were included in all model fits. A visual depiction of predator catch at various repair types can be found in Figure 15.

The final linear mixed model for evaluation predator abundance with respect to Mitigated Repair Site Types was in the form

$$\log(c) = \beta_0 + \beta_{1:2}RegionA:B + \beta_{3:4}Event_{Mid:Late} + \beta_{5:7}TypeA:C + Region|Event$$

where c is predicted predator abundance (black bass or combined black bass/striped bass/pikeminnow), Regions A and B correspond to regions Garcia Bend (RM40-70) and Knights Landing (RM70-95), respectively (the reference region for this model was Ko-Ket [RM21-39]), *Event* represents the sampling time, *Region|Event* is an interaction term, and Types A through C correspond to Mitigated Repair Site Types 10:1 Slope/Bench, No Bench, and Wetland Trench, respectively.

In the black bass model, river region and timing of sampling had a significant effect on bass abundance, which was lower in upstream areas (Knights Landing Region), and increased through time (more bass captured during later sampling events; Table 10). These findings suggest that bass abundance is generally higher closer to the estuary, but also indicates a temporal trend in bass abundance. The temporal abundance increase could be attributed to seasonal shifts in habitat usage (either upstream movements or movements from deep water habitat to nearshore areas sampled effectively by electrofishing), yet details on habitat usage in areas outside the river margins along monitoring sites are unknown, preventing more detailed conclusions.

For both predator categories, abundance was generally higher at Mitigated Repair Sites when compared to naturalized locations. The most pronounced positive association, regardless of predator category, was found at Wetland Trench Sites.

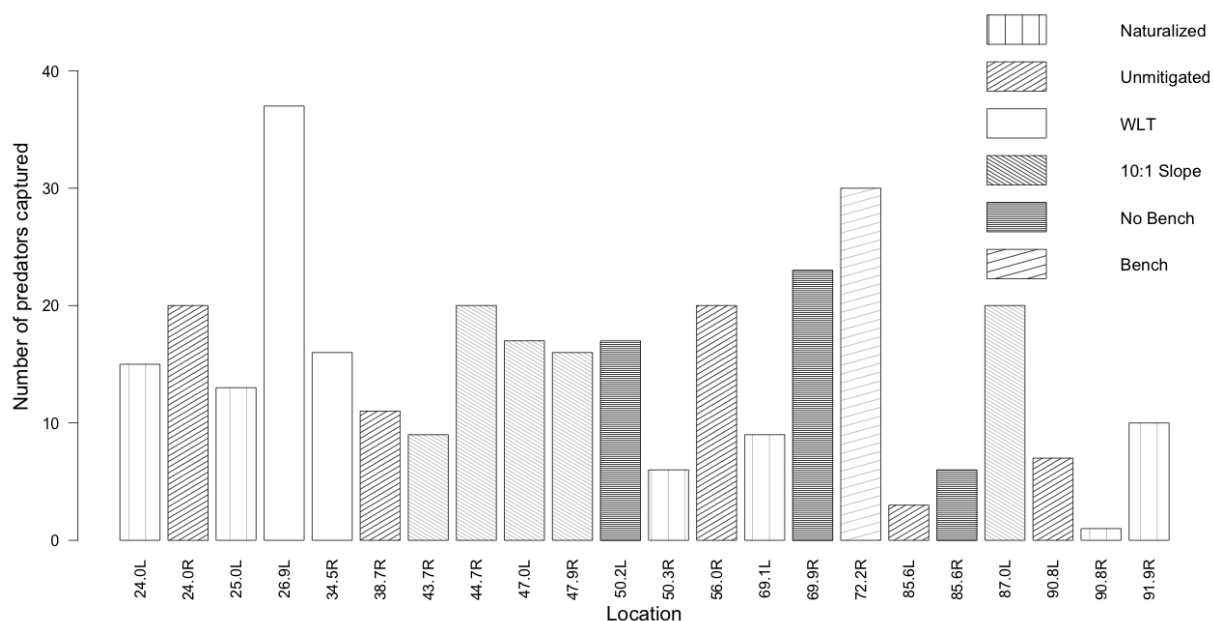


Figure 15. River Mile, Site Type, and Number of Predators captured during the 2012 electrofishing surveys.

Table 10. Influence of Mitigated Repair Site Type on abundance of black bass and other piscivores sampled in March and April 2012.

Variable (Reference)	Category	<u>Black Bass</u>			<u>All Predators</u>		
		Coefficient	SE	Sig.	Coefficient	SE	Sig. ¹
Region (Ko-Ket)	Garcia Bend	-0.42	0.40	NS	-0.29	0.25	NS
	Knights Landing	-1.61	0.75	*	-0.53	0.37	NS
Event (First)	Middle	0.20	0.22	*	0.34	0.16	*
	Last	0.62	0.20	**	0.58	0.15	***
Repair Type	10:1 Slope/ Bench	0.66	0.34	NS	0.61	0.20	**
	No Bench	1.05	0.35	**	0.61	0.22	**
	WLT	1.37	0.28	***	0.77	0.21	***

¹Significance categories: “NS” (not significant), “*” (p<0.05), “**” (p<0.01), and “***” (p<0.001)

Habitat Features

To investigate potential relationships between predator abundance and habitat variables, the model used in this step included the following habitat features (and spatial and temporal variables included in all models): shade (present/absent), density of emergent vegetation (absent/low/medium/high, absent was used as the reference category), density of emergent woody material (absent/low/medium/high, absent was used as the reference category), flow (feet/s), depth (feet), terrestrial vegetation (percent bank coverage) and terrestrial woody material (percent bank coverage).

The predictive equations for evaluating the effect of habitat features on predator abundance was in the general form

$$\log(c) = \beta_0 + \beta_1 \text{Shade}_{\text{present}} + \beta_{2:4} \text{EV}_{\text{dens.}} + \beta_{5:7} \text{EW}_{\text{dens.}} + \beta_8 \text{Flow} + \dots$$

$$\beta_9 \text{Depth} + \beta_{10} \text{TV}\% + \beta_{11} \text{TW}\%$$

where c is predicted predator abundance (black bass or combined black bass/striped bass/pikeminnow), *Shade* refers to shade presence, *EV* and *EW* correspond to the density of emergent vegetation and emergent woody material, respectively (low, medium and high, absence is used as the reference category), *TV* and *TW* refer to the percent coverage of terrestrial vegetation and terrestrial woody material, respectively.

In this analysis, flow and depth significantly influenced bass abundance. Black bass appear more likely to inhabit shallow areas (negative correlation to depth) and locations where some current is present. In addition, fewer black bass were observed in areas where the density of emergent vegetation was high, yet no significant correlation was apparent between bass abundance and intermediate and low levels of emergent vegetation (Table 11). This may seem counterintuitive, as bass are well known to inhabit areas where aquatic vegetation is abundance, which they use for shelter and cover. It is possible that dense vegetation, when present, partially obstructed stunned fish and led to negatively biased catch rates at these locations.

In the model used to evaluate combined predator abundance (black bass, striped bass and Sacramento pikeminnow), none of the microhabitat features examined appeared to have a significant effect on predator abundance, with the exception of high density emergent woody material which, similar to high vegetation density in the bass model, appeared to negatively impact predator abundance, while emergent woody material at low and medium densities did not have a significant effect (Table 11).

These findings contrast slightly with those from the 2011 monitoring, during which none of the microhabitat features appeared to have a significant impact on predator abundance, and Mitigated Repair Site Type was the sole variable influencing abundance of piscivorous species.

Table 4. Influence of microhabitat features on abundance of piscivorous predators sampled in March and April 2012

Habitat Variable	Levels	Black Bass			All Predators		
		Coefficient	SE	Sig.	Coefficient	SE	Sig.
Shade	-	0.01	0.19	NS	-0.17	0.14	NS
Flow	-	1.54	0.50	**	0.48	0.35	NS
Depth	-	-0.13	0.07	*	-0.01	0.04	NS
Terrestrial Veg.	-	0.03	0.65	NS	0.35	0.53	NS
Terrestrial WM	-	0.53	0.71	NS	0.23	0.57	NS
Emergent Veg.	Low	-0.18	0.21	NS	-0.10	0.16	NS
	Medium	-0.12	0.26	NS	0.00	0.20	NS
	High	-0.80	0.35	*	-0.10	0.23	NS
Emergent WM	Low	-0.05	0.19	NS	0.01	0.14	NS
	Medium	-0.34	0.32	NS	0.01	0.22	NS
	High	-15.86	405.3	NS	-1.14	0.44	**

¹ Significance categories: "NS" (not significant), "*" (p<0.05), "**" (p<0.01), and "***" (p<0.001)

Combined Mitigated Repair Site Types/Habitat Features

Most of the habitat variables evaluated in the preceding model did not show a significant influence on predator abundance and were excluded from further analysis. Those variables that did have a significant effect on predator abundance in the previous analysis step were included in a model incorporating effects of both Mitigated Repair Site Type and Habitat Features. Note that the significance level of some variables differed between groups (bass only vs. all predators, see Tables 10 and 11), and combined model fits differ slightly between predator categories (e.g., flow and depth were included in the bass model, but not in the combined predator model). In general, the combined model was in the form

$$\log(c) = \beta_0 + \beta_{1:2}RegionA:B + \beta_{3:4}Event_{Mid:Late} + \beta_{5:7}TypeA:C + \dots$$

$$(\beta_x Flow + \beta_y Depth/h) + (\beta_{z:z+2}EV_{dens.}) + (\beta_{z:z+2}EW_{dens.})$$

Of the microhabitat variables considered significant in predicting bass abundance, only high density emergent vegetation was still found to be significant in predicting bass abundance in the combined model, while flow and depth did not appear to affect abundance after accounting for Mitigated Repair Site Type (Table 12).

Habitat features included in evaluation combined predator abundance in the combined model (emergent woody material) were no longer considered significant after accounting for the effects of design type and other covariates included in the model fits (river region and sampling event).

All repair types were associated with significantly higher abundance of black bass and salmonid predators in general (as compared to naturalized sites, used as the reference category). The magnitude of the coefficients indicates that highest abundance of bass and combined salmonid predators was found at Wetland Trench Sites, followed by No Bench, and 10:1 Slope/Bench Sites.

Table 5. Influence of Mitigated Repair Site (and microhabitat variables on abundance of piscivorous species sampled during in March and April 2012.

Variable	Category	<u>Black Bass</u>			<u>All Predators</u>		
		Coefficient	SE	Sig.	Coefficient	SE	Sig. ²
Region (<i>Ko-Ket</i>)	Garcia Bend	-0.79	0.55	NS	-0.21	0.26	NS
	Knights Landing	-1.60	0.64	*	-0.37	0.29	NS
Event (<i>First</i>)	Middle	0.00	0.22	NS	0.29	0.15	NS
	Last	0.34	0.21	NS	0.49	0.15	***
Repair Type	10:1 Slope/ Bench	0.82	0.42	*	0.53	0.23	*
	No Bench	1.14	0.40	**	0.53	0.25	*
	WLT	1.16	0.34	***	0.81	0.22	***
Flow ¹	-	0.51	0.52	NS	-	-	-
Depth ¹	-	0.02	0.10	NS	-	-	-
Emergent Veg. ¹	Low	-0.12	0.21	NS	-	-	-
	Medium	-0.29	0.26	NS	-	-	-
	High	-0.74	0.33	*	-	-	-
Emergent WM ¹	Low	-	-	-	-0.11	0.14	NS
	Medium	-	-	-	0.22	0.21	NS
	High	-	-	-	-0.79	0.41	NS

¹ Variable only included in model fit if deemed significant in previous analysis steps.

² Significance categories: “NS” (not significant), “*” (p<0.05), “**” (p<0.01), and “***” (p<0.001)

6.1.3 Evaluation of fish habitat associations based on cumulative monitoring data

Site Types

As described in detail in preceding sections, the first analysis step incorporated the effect of Site Type (Mitigated Repair, Unmitigated Repair, and Naturalized) to determine whether different types of repairs (i.e., Unmitigated versus Mitigated Repairs) influence abundance of Chinook salmon fry and juveniles in comparison to each other and with Naturalized Sites.

Because two years of data were combined for this analysis, “Year” and the interaction term between Year and sampling event was included to account for catch differences attributable to temporal differences in abundance. Though flow was somewhat variable during sampling events in 2011 (wet water year) and 2012 (below normal water year), and ranged from nearly 13,000 cfs

to 17,000 cfs (mean flow during 3-4 day sampling event), discharge was not included as an explicit variable in the statistical analysis but accounted for by including the “Event” variable. While the interaction between flow and season may influence in-river distribution of juvenile salmonids and outmigration timing, the focus of this monitoring effort is to evaluate site- and repair-type specific habitat use. Point-by-point review of video footage captured during electrofishing allows for accurate incorporation of certain habitat features and their availability to fish (which changes with inundation level), and is considered to adequately incorporate the effects of changing conditions on fish habitat use.

Coefficients for the different Sites Types (regardless of the reference category used in the model) indicate that abundance of Chinook salmon fry was lowest at unmitigated repair sites, while there was no significant difference in fry abundance between Naturalized and Mitigated Repair Sites. Juvenile abundance was significantly higher at both, Naturalized and Mitigated Repair Sites compared to Unmitigated Sites (Table 13), and the highest abundance was observed at Naturalized Sites. Data from Unmitigated Repair Sites was excluded from further analysis to focus on the parameters of main interest in this study.

Table 6. Influence of combined Site Type on abundance of Chinook salmon and piscivorous species sampled in 2011 and 2012.

Reference Category	Site Type	<u>Fry</u>			<u>Juvenile</u>			<u>Predators</u>		
		Coef.	SE	Sig. ¹	Coef.	SE	Sig. ¹	Coef.	SE	Sig. ¹
Unmitigated	Mitigated	0.93	0.20	***	0.66	0.09	***	-0.01	0.11	NS
	Naturalized	0.99	0.20	***	0.99	0.09	***	-0.65	0.13	***
Mitigated Repair	Unmitigated	-0.93	0.20	***	-0.66	0.09	***	0.01	0.11	NS
	Naturalized	0.06	0.12	NS	0.33	0.06	***	-0.64	0.12	***
Naturalized	Unmitigated	-0.99	0.20	***	-0.99	0.09	***	0.65	0.13	***
	Mitigated	-0.06	0.12	NS	-0.33	0.06	***	0.64	0.12	***

¹ Significance categories: “NS” (not significant), “*” (p<0.05), “**” (p<0.01), and “***” (p<0.001)

Mitigated Repair Site Types

GLMMs were used to assess whether individual Mitigated Repair Site Types (10:1 Slope/Bench, No Bench, Wetland Trench) influence Chinook salmon fry and/or juvenile abundance based on cumulative data from 2011 and 2012 monitoring. The variable “Year” and the interaction term between year and sampling event was retained to account for differences resulting from variable inter-annual abundance. Naturalized Sites were used as the reference category in all model fits.

Fry

In this model, no significant abundance difference between natural sites and the 10:1 Slope/Bench or Wetland Trench Site Type could be discerned; however, Chinook salmon fry abundance was significantly lower at No Bench Sites when compared to Naturalized Sites. These results are consistent with those based on analysis of single-year (2011) survey data.

Juveniles

Similar to results based on analysis of single-year (2012) data, highest abundance of juvenile Chinook salmon is found at 10:1 Slope/Bench Sites, followed by Naturalized Sites. Reduced

abundance was observed at Wetland Trench Sites (compared to Naturalized Sites; consistent with monitoring results from 2011), and no discernible abundance differences were found between Naturalized and No Bench Sites during analysis of the combined data set (Table 14).

Predators

Based on analysis of the combined dataset, predator abundance is higher at all Mitigated Repair Sites than at Naturalized Sites, yet among repair sites, predators abundance is lowest at 10:1 Slope/Bench sites, the design type occupied by higher abundances of juvenile Chinook salmon. Highest predator abundance, as documented previously based on single-year analysis in 2011 and 2012, was found at Wetland Trench sites, those locations where Chinook salmon juveniles were least likely to be observed during field sampling. It is unclear whether this is a result of predator-prey interaction or attributable to differences in habitat preference between juvenile Chinook salmon and piscivores. Intermediate predation risk (based on predator abundance) was found at No Bench Repair sites.

Table 7. Influence of Mitigated Repair Site Type on abundance of Chinook salmon fry and juveniles and predatory fishes sampled in 2011 and 2012.

Reference Category	Site Type	<u>Fry</u>			<u>Juvenile</u>			<u>Predators</u>		
		Coef.	SE	Sig. ¹	Coef.	SE	Sig. ¹	Coef.	SE	Sig. ¹
Naturalized	10:1 Slope/Bench	0.31	0.16	NS	0.65	0.11	***	0.51	0.17	**
	No Bench	-1.48	0.33	***	0.05	0.13	NS	0.59	0.18	**
	WLT	-0.11	0.18	NS	-0.95	0.09	***	0.82	0.19	***

¹Significance categories: “NS” (not significant), “*” (p<0.05), “**” (p<0.01), and “***” (p<0.001)

Habitat Features

To investigate potential relationships between Chinook salmon fry and juvenile abundance and various microhabitat features, the model used in this step included the following habitat variables (in addition to the spatial and temporal variables included in all models, namely Year, Event, Region and their interaction terms): shade (present/absent), density of emergent vegetation (absent/low/medium/high, absent was used as the reference category), density of emergent woody material (absent/low/medium/high, absent was used as the reference category), flow (feet/s), depth (feet), bank slope, current gradient, terrestrial vegetation (TV, percent bank coverage) and terrestrial woody material (percent bank coverage).

Fry

Of the habitat variables analyzed in this model, only average depth, level of bank coverage of vegetation and woody material and density of instream woody material appeared to have a significant effect on the abundance of Chinook salmon fry (Table 15).

As indicated by the negative coefficient, fry abundance was negatively associated with depth, which indicates that more fry were encountered at sites with shallow water than at those with a deeper average depth. This result stands in contrast to the findings from the single year analysis

(2012), which indicated higher abundance of Chinook salmon fry in deep-water habitat. As discussed earlier, the positive association between fry abundance and deep-water in 2012 was deemed inconclusive and contradictory to broadly acknowledged preference for shallow water habitat by this life history stage. The conclusions resulting from analysis of the combined data set (2011 and 2012) support the notion that fry do prefer shallow water habitat, and appear to confirm that contradictory results based on single year analysis is a function of limited sample size in 2012.

While emergent vegetation did not appear to affect fry abundance at low and medium densities, fewer fry were observed at location where density of emergent vegetation was high. It is possible that high densities of emergent vegetation visually obstruct fry or prevent them from being drawn to the boat's electrodes, which may reduce capture efficiency (in particular in turbid water) and, subsequently, limit the statistical significance of this explanatory variable. None of the other microhabitat variables (shade, flow, slope, current gradient, density of woody material) had a significant effect on fry abundance in this model.

Juveniles

Shade, flow, and emergent woody material (all densities) were negatively associated with the abundance of juvenile Chinook salmon (Table 15). Positive associations were found between juvenile abundance and terrestrial woody material and emergent vegetation. The negative coefficient for flow indicates that juvenile Chinook salmon were associated more often in areas with slow current. Elevated densities of juvenile Chinook salmon in areas with slow velocity habitat and emergent (or submerged) vegetation indicate that juveniles were using habitat with characteristics similar to those attributed to floodplains often considered beneficial for salmonid rearing. In contrast, the inverse association between juvenile abundance and current was found in 2011, when juvenile Chinook salmon abundance was higher in areas with faster current velocities. The presence of shade also had a negative effect on the abundance of juvenile Chinook salmon in 2012; a factor that did not appear to have a significant effect on habitat use in single year analysis for 2011. Percent bank coverage of terrestrial woody material at the sampling site appeared to have a significant positive effect on the abundance of Chinook salmon juveniles (Table 15). Though these terrestrial habitat features appear to influence Chinook salmon abundance, the mechanisms of this interaction are unclear. It is possible that these features affect variables not accounted for in the models presented here (such as prey supplementation from adjacent vegetation), or are reflective of associated habitat conditions below the water surface level (diverse habitat on shore continues under water). A positive association between emergent vegetation and juvenile Chinook salmon abundance supports the latter explanation. Juvenile Chinook salmon abundance was lower in areas of emergent woody material in 2012 (Table 9).

Predators

The only habitat features that appear to affect predator abundance are bank slope (positive coefficient) and current gradient (negative coefficient), indicating that piscivorous species encountered during the surveys were associated more often with areas having steep slopes and relatively constant current profiles. No other habitat variables appeared to significantly affect predator abundance (Table 15).

Table 8. Influence of microhabitat features on abundance of Chinook salmon and salmonid predators sampled in 2011 and 2012.

Habitat Variable	Levels	<u>Fry</u>			<u>Juvenile</u>			<u>Predators</u>		
		Coef.	SE	Sig. ¹	Coef.	SE	Sig. ¹	Coef.	SE	Sig. ¹
Shade	-	-0.13	0.14	NS	-0.20	0.06	**	-0.18	0.12	NS
Flow	-	2.15	1.90	NS	-2.99	1.20	*	3.49	1.71	NS
Depth	-	-1.19	0.43	**	0.46	0.25	NS	-0.40	0.34	NS
Terr. Veg.	-	-1.45	0.27	***	0.34	0.20	NS	0.09	0.29	NS
Terr. WM	-	-1.23	0.29	***	0.45	0.22	*	-0.02	0.33	NS
Slope	-	3.66	4.68	NS	-4.64	2.66	NS	9.51	3.75	*
Current gradient	-	-18.51	19.34	NS	23.22	12.01	NS	-35.86	16.97	*
Emergent Veg.	Low	0.07	0.14	NS	0.34	0.08	***	0.13	0.14	NS
	Medium	0.17	0.16	NS	0.27	0.10	**	0.19	0.17	NS
	High	-0.51	0.22	*	0.24	0.10	*	0.26	0.19	NS
Emergent WM	Low	-0.01	0.15	NS	-0.30	0.07	***	-0.14	0.12	NS
	Medium	0.06	0.19	NS	-0.27	0.10	**	-0.01	0.18	NS
	High	-0.04	0.23	NS	-0.41	0.14	**	-0.45	0.26	NS

¹ Significance categories: “NS” (not significant), “*” (p<0.05), “**” (p<0.01), and “***” (p<0.001)

Combined Mitigated Repair Site Types/Habitat Features

To investigate potential relationships between Chinook salmon fry and juvenile abundance with the Mitigated Repair Site Types and habitat features, models that combined these variables were used in this step. Variables included the different Mitigated Repair Site Types (10:1 Slope/Bench, No Bench, Wetland Trench; with Naturalized Sites used as the reference category) and those habitat features that were previously determined to have a significant effect on fish abundance (Tables 14 and 15), in addition to the spatial and temporal variables included in all models.

Fry

Based on the combined model using data from 2011 and 2012, Chinook salmon fry were more abundant in Naturalized Sites versus Mitigated Repair Sites. While no significant difference in fry abundance could be attributed to Wetland Trench Sites, reduced densities were documented at 10:1 Slope/Bench Sites and No Bench Sites, the latter being associated with the lowest fry abundance (Table 16). Interestingly, reduced fry densities were also associated with bank coverage of terrestrial vegetation and woody material, as well as with high density emergent vegetation. While this may seem to stand in contrast to the notion that fry often use habitats where vegetation abounds, it may be an indication of aversion to high density vegetation rather than vegetated habitat in general. In addition, dense vegetation may have reduced the capture success for fry, which could have led to underestimated fry abundance and affected subsequent model fitting.

None of the other microhabitat variables (shade, flow, slope, current gradient, density of woody material) had a significant effect on fry abundance in this model.

Juveniles

Chinook salmon juveniles were significantly more abundant at 10:1 Slope/Bench Sites and significantly less abundant at Wetland Trench Sites, but no difference in abundance of Chinook salmon juveniles was found between Naturalized Sites and No Bench Sites. Flow was significantly associated with the abundance of juvenile Chinook salmon (Table 9). The negative coefficient indicates that juvenile Chinook salmon were more abundant in habitats with slow current. The inverse association between juvenile abundance and current, though not apparent in 2011 when juvenile Chinook salmon were associated with faster current velocities, was significant in the combined dataset. The presence of shade also had a negative effect on the abundance of juvenile Chinook salmon in the long-term dataset, a factor that did not appear to have a significant effect on habitat use in the previous year. Percent bank coverage of terrestrial woody material at the sampling site also had a significant positive effect on the abundance of Chinook salmon juveniles (Table 16); whereas, juvenile abundance decreased incrementally⁵ with in-water woody material density (an association that did not become apparent during analysis of single year survey data). In contrast, all densities of emergent vegetation were associated with increased juvenile abundance.

Predators

Similar to the results from previous years and those based on analysis of single-year data sets, none of the microhabitat variables incorporated in the models appeared to have significant effects on predator abundance after accounting for effects of site type/repair design. Notably, predator abundance was higher at all Mitigated Repair Site Types than at Naturalized Sites, and not significantly different from Unmitigated Repair Sites consisting mainly of rock revetment.

⁵ Densities of woody material and vegetation were defined as categorical (absent, low, medium, high), not continuous, variables.

Table 9. Influence of Mitigated Repair and microhabitat variables on abundance of Chinook salmon fry and juvenile and salmonid predators sampled in 2011 and 2012.

Habitat Variable	Levels	Fry			Juvenile			Predators		
		Coef.	SE	Sig. ²	Coef.	SE	Sig. ²	Coef.	SE	Sig. ²
Mitigated Repair Site	10:1 Slope/ Bench	-0.52	0.25	*	0.35	0.11	**	0.73	0.22	***
	No Bench	-2.20	0.35	***	-0.21	0.14	NS	0.75	0.20	***
	WLT	-0.33	0.20	NS	-0.90	0.10	***	0.73	0.20	***
Shade ¹	-	-	-	-	-0.16	0.06	**	-	-	-
Flow ¹	-	-	-	-	-0.65	0.19	***	-	-	-
Depth ¹	-	-0.19	0.08	*	-	-	-	-	-	-
Slope ¹	-	-	-	-	-	-	-	1.03	0.62	NS
Current gradient ¹	-	-	-	-	-	-	-	-1.96	2.90	NS
Emergent WM ¹	Low	-	-	-	-0.28	0.07	***	-	-	-
	Medium	-	-	-	-0.35	0.10	***	-	-	-
	High	-	-	-	-0.55	0.14	***	-	-	-
Emergent Veg. ¹	Low	0.08	0.15	NS	0.41	0.08	***	-	-	-
	Medium	0.24	0.17	NS	0.38	0.10	***	-	-	-
	High	-0.44	0.21	*	0.25	0.10	***	-	-	-
Terr. Veg. ¹	-	-1.98	0.25	***	-	-	-	-	-	-
Terr. WM ¹	-	-1.87	0.27	***	-0.04	0.13	NS	-	-	-

¹ Variables identified by superscript are not included in the combined model for abundance due to lack of significance in previous analysis steps

² Significance categories: “NS” (not significant), “*” (p<0.05), “**” (p<0.01), and “***” (p<0.001)

6.2 Sampling with Alternative Collection Methods (Fish Objective 3)

Prepositioned area electrofishing (PPAE)

Catches remained low throughout this survey, presumably due to a combination of low abundance of fish and gear limitations that became apparent during testing. While changing the electric output (from a low range of 40% at 60 DC to a high range of 40% 60 AC) resulted in slightly increased catches (several juvenile Centrarchids and Catastomids were captured), the overall performance of this technique is difficult to evaluate as actual fish densities in our sampling locations was unknown. No differences between the two design types were noted.

Observations of juvenile Sacramento sucker and red shiner within the activated grid in shallow water indicate that the grids were functional; however, these fish were generally able to successfully evade capture by moving to and remaining in the center of the grid, only being immobilized when they attempt to cross the electrodes. Increased capture success was achieved by reducing the distance between the two electrodes; though this increased mortality of affected individuals (cumulative observed mortality for all tests amounted to approximately 30 juvenile red shiner). However, as small individuals are much less susceptible to capture by electrofishing, it remains unclear how larger individuals are affected by their relative position within the grid.

While PPAE holds promise for sampling locations that are otherwise difficult to access (due to dense vegetation/woody material, see Figure 16), one of its main limitations is visibility: as the grid is positioned on the riverbed, fish present within the area of influence of the electric field are drawn towards the electrodes away from the water surface. In contrast to boat electrofishing where fish are drawn towards the electrodes suspended from beams, PPAE prevents sampling personnel from effectively capturing stunned fish if water depths exceeds visibility, limiting the use of PPAE to shallow habitats or less turbid water. As removing and deploying of these grids is time consuming (especially considering the required resting period between deployment and activation), allowing, for a given amount of time/personnel hours, sampling only at a very limited number of locations. The resulting low number of replicates prevent or severely limit the application of this to large scale monitoring efforts.



Figure 16. Deployment of prepositioned electrofishing grid in dense structure.

However, its applicability for studies that require repeated sampling of geographically limited target areas that allow the grids to remain deployed between sampling events should be considered. Prior to such implementation, detailed current profiles within the grid should be established under laboratory conditions to better understand the effects of power settings on the strength and extent of the electric field. This would serve to maximize sampling efficiency while limiting excessive mortality of sampled individuals.

Breder traps

No fish were captured in Breder traps. As this method is applicable for sampling larval and juvenile fish, lack of fish captures may be attributed to low abundance of juveniles during the time of sampling.



Figure 17. Periodic checking of Breder traps. No fish were captured in these traps.

Quatrefoil light traps

No fish were captured in the quatrefoil traps during the first three trapping intervals (checking light traps, Fig. 18). The fourth trapping interval resulted in catches of two juvenile inland silverside in each trap.

As expected, light traps proved most effective after nightfall. Low overall capture rates may be attributed to low seasonal abundance of juvenile and larval fish. Trapping was discontinued after the fourth trapping interval, since we determined that these traps were able to collect larval and juvenile fish associated with targeted habitat features.

These traps are a promising tool for sampling abundances of larval and juvenile fish that are not susceptible to capture by other sampling techniques (e.g., electrofishing), and can provide abundance indices for areas with dense cover (e.g. Fig. 19), as are often encountered along the bank of the Sacramento River.



Figure 18. Checking quatrefoil light traps as part of evaluation of supplementary collection methods.



Figure 19. Quatrefoil light trap deployed in shallow water near dense cover.

6.3 Habitat Monitoring (Habitat Objective 1)

In 2012, SAM habitat surveys were conducted by the USACE (in prep).

6.3.1 Photo Point Monitoring

Steep banks and dense vegetation encountered at many SRBPP sites made traditional photo point monitoring impractical. Furthermore, channel widths encountered were typically too wide to permit establishment of photo points on opposing banks due to equipment focal limitations. At some sites, the orientation of the sun created glare that necessitated boat positioning in a manner other than previously described, generally closer to either the upstream or downstream end of the project site. Photographs taken prior to the 2012 monitoring events included shore-based photos and photos taken mid-channel from boat. No shore-based photos were taken at photopoints previously established in 2011 because unmitigated views are now obstructed due to dense riparian willow/cottonwood stands that have become established at most sites. Photos taken from mid-channel can be used to provide an indication of habitat changes over time and are repeatable since views will not become obstructed by vegetation changes over time. Flows listed in photo captions are from the closest gauge at Wilkens Slough (CDEC: WLK, RM 117.6) and do not necessarily reflect actual flows present at project sites downstream.

As panoramic images lose their resolution and are difficult to view in a text document (e.g., Figure 20), digital copies of these images are supplied on CD. At one site (26.0L), poor lighting conditions from all angles necessitated taking several overlapping close-up images while floating through the site. Due to the moving perspective, images for site 26.0L do not lend themselves to being stitched into a single panoramic photo and are provided as separate photos.



Figure 20. An example of a project site (91.9R) where a panoramic photo was produced from several consecutive images taken from a boat positioned midriver.

6.3.2 Sidescan Sonar

In contrast to the results described in Kaeser and Litt (2010), imagery produced in 2012 generally lacked the necessary resolution to consistently identify IWM. During each of the two survey events, methodology was adjusted to better refine the detail of the imagery produced. One of the primary challenges associated with the ability to identify IWM was attributed to the general size and orientation of the IWM present at project sites. Kaeser and Litt (2010) used their SSS to identify large historically logged old growth timber that sank in random locations

during transport downstream. The smaller size, occasional high density, and close proximity to shore of IWM within the SRBPP survey sites provided challenges that were not described in, or encountered by Kaeser and Litt (2010).

In an effort to improve imagery, the range of the SSS was reduced and the boat was positioned closer to shore (approximately 50 feet/15 meters). Closer proximity to shore, in conjunction with nearshore obstacles, increased the precision required to properly position the boat as it moved through the target area. The highest quality SSS imagery was produced when the boat was able to track perfectly parallel to the shoreline at a constant speed (Figures 21 and 22); whenever this was not possible, the resulting image distortion could not be corrected during post processing. Furthermore, waves created by other boats or wind caused the boat and transducer to plunge up and down vertically in the water column, rendering resulting images unusable.

Due to inconsistent water velocity/direction created by eddies and channel morphology, it often proved difficult to keep the boat at a constant speed and parallel to the shoreline, which resulted in poor quality SSS. The use of a drift sock or bucket was employed to aid in more consistent boat tracking; however, this approach was not adequate in all situations. One potential solution to mitigate inconsistencies in boat tracking (not tested in 2012) would be to configure the SSS transducer and GPS antenna to a tiller type mount (similar to an outboard boat motor). This would allow the unit to be independently controlled and remain in a more parallel orientation to shore regardless of boat position.

At sites where SSS images were adequate to allow IWM delineation, catch from electrofishing surveys was overlaid to look for associated underwater habitat. However, it was discovered that the differences in GPS and to overlay positions for the electrofishing video did not provide the necessary resolution to make any definitive associations to habitat. In the future, both GPS units would need to have sub-meter, survey grade resolution to allow any such associations.

Potential future improvements to SSS methodology include

- Use of a swiveling transducer mount that will facilitate maintaining a parallel orientation to the shoreline irrespective of boat orientation
- Use of sub-meter GPS devices with identical resolution for the habitat scanning and electrofishing effort

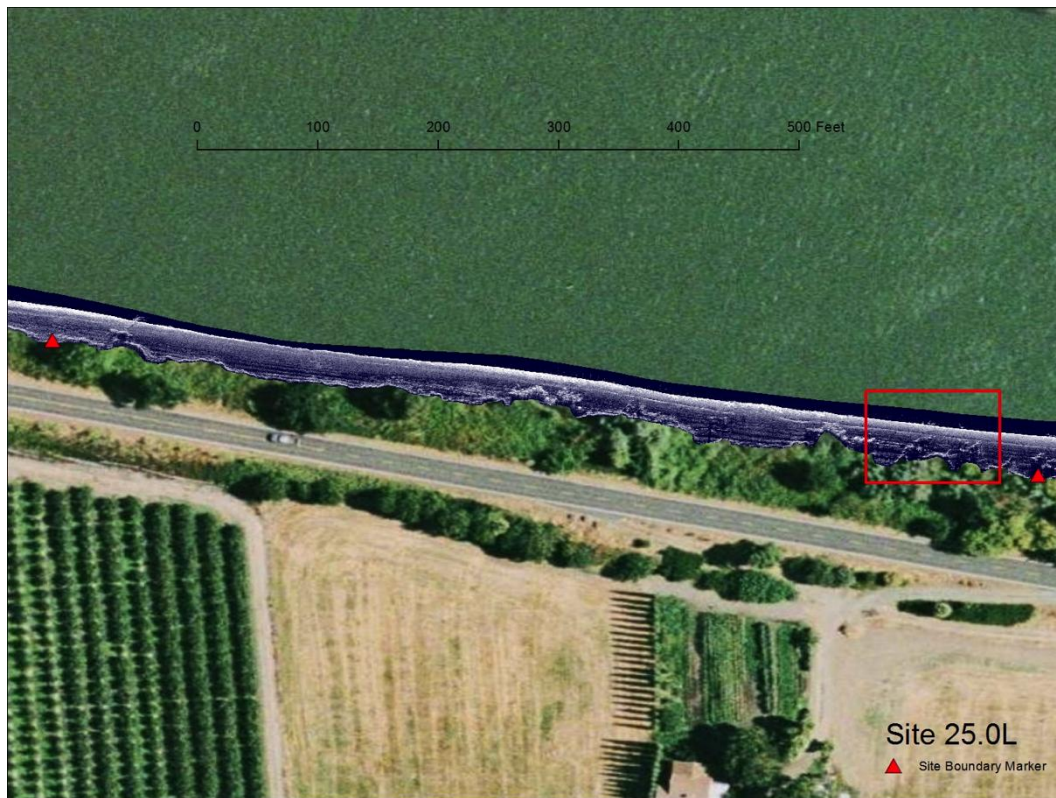


Figure 21. Overlay of aerial and sidescan sonar image at Site 25.0L (Naturalized). Red box indicates area illustrated in more detail in Figure 22.

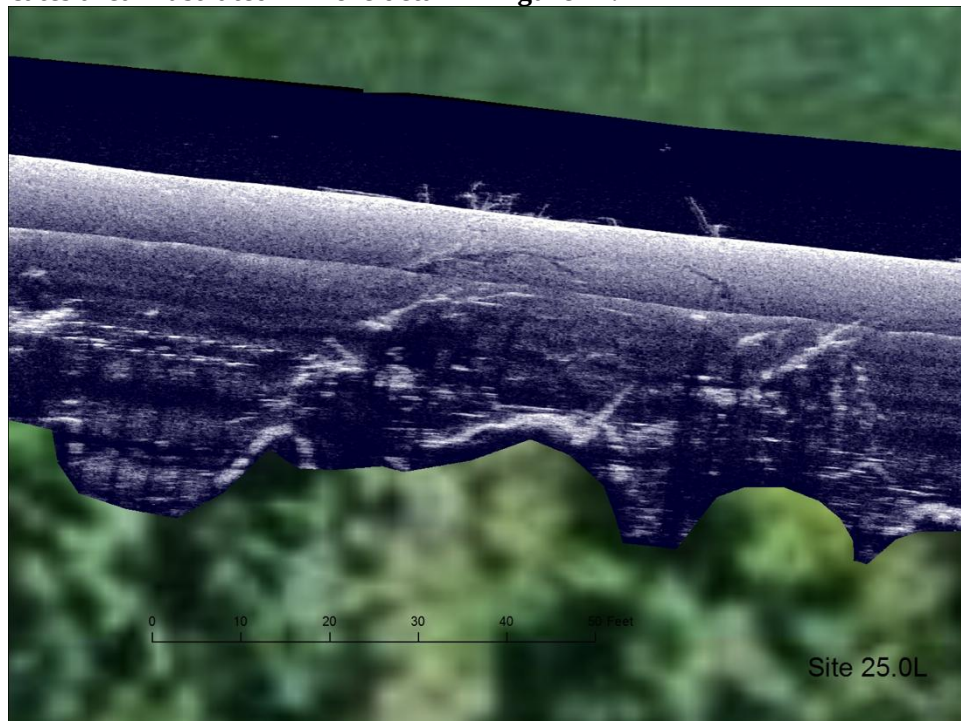


Figure 22. Sidescan sonar image from Site 25.0L, illustrating naturally deposited submersed woody material.

7.0 CONCLUSIONS

2012 Monitoring

Fish communities were relatively similar between Naturalized Sites, and Mitigated Repair Sites and Reference Sites. Based on electrofishing catch, fish community abundance and composition were similar between different site categories. Similarly, a large degree of overlap in species composition was found between Naturalized Sites and Mitigated Repair Sites (10:1 Slope, Bench, No Bench, Wetland Trench; Morisita-Horn indices greater than 0.87 for all pairwise comparisons). These results were mainly driven by the proportions of Chinook salmon and black bass encountered at these sites, which outweighed site-specific differences in the presence of less abundant species.

Juvenile Chinook salmon, but not fry, were encountered at elevated densities at Mitigated Repair Sites compared to Unmitigated Repair Sites. Based on general linear model (GLM) analysis, the abundance of Chinook salmon juveniles is significantly higher at the Mitigated Repair Sites compared to Unmitigated Repair Sites, Chinook salmon appear to prefer Mitigated Repair Sites containing designed habitat features compared to Unmitigated Repair Sites consisting primarily of rock revetment. However, Chinook fry abundance was not significantly different between Unmitigated Repair Sites and Mitigated Repair Sites, though Naturalized Sites were preferred over Unmitigated Repair Sites.

Based on electrofishing, Chinook salmon juvenile abundance is higher at 10:1 Slope/Bench Sites than both Naturalized Sites and other Mitigated Repair Site Types. Among Mitigated Repair Sites surveyed in 2012, Chinook salmon juveniles were strongly associated with 10:1 Slope/Bench Site Types. For Chinook salmon fry, no significant abundance differences were found between Naturalized, Wetland Trench, and 10:1 Slope/Bench Sites, though significantly reduced densities were observed at No Bench Sites. Wetland Trench Sites were associated with Lower abundance of juveniles in 2012.

Regardless of Mitigated Repair Site Type, Chinook salmon fry and juveniles are more commonly found in areas with reduced current velocity, and juveniles are found in shallow water areas. Chinook salmon juveniles appear to prefer shallow water habitat, and flow was negatively correlated to fry and juvenile abundance in 2012. This suggests that slow velocity areas are important for Chinook salmon rearing and refuge.

Chinook salmon fry abundance decreased as bank coverage of terrestrial vegetation and terrestrial woody material increased. Based on electrofishing, fry abundance decreased as bank coverage of terrestrial vegetation and terrestrial woody material increased. Though these terrestrial habitat features appear to positively impact *juvenile* abundance, the mechanisms of this interaction are unclear. It is possible that these features are reflective of associated habitat conditions below the water surface level, which is supported by observed positive association between juvenile abundance and low to medium density emergent vegetation.

Predation risk to Chinook salmon fry and juveniles is highest at Wetland Trench Sites, and predator abundance was higher at all Mitigated Repair Sites than at Naturalized Sites. Analysis of catch data of potential salmonid predators (black bass, striped bass, and Sacramento pikeminnow) indicates that combined predator abundance significantly higher at Mitigated

Repair Sites than at Naturalized Sites, and that the highest predator density was associated with Wetland Trench sites, followed by No Bench and 10:1 Slope/Bench locations. These results remain consistent for combined survey data (2011 and 2012).

The overall effects of Mitigated Repair Site designs appear to outweigh the importance of individual microhabitat features in determining predator abundance.

None of the habitat variables analyzed in this study appeared to be consistently associated with predator abundance. More detailed habitat typing (at each point for which catch data is available) rather than using mean values for these variables across each site may provide additional resolution in the future.

Long-term Monitoring

Regardless of Repair Site Type, Chinook salmon fry and juveniles are more commonly found in areas with reduced current velocity, and juveniles are found in shallow water areas. Consistent with results from previous years, Chinook salmon juveniles appear to prefer shallow water habitat, though H.T. Harvey & Associates and PRBO (2011) indicated there was insufficient evidence to conclude a significant effect of water depth during 2009 and 2010. Current velocity was not included in model fitting in 2009 and 2010, and while juvenile Chinook salmon appeared to preferentially inhabit areas with increased current velocity in 2011, flow was negatively correlated to fry and juvenile abundance in 2012. This suggests that slow velocity areas are important for Chinook salmon rearing and refuge, a result that was emphasized by significant association between abundance of juveniles and reduced current in analysis of combined monitoring data (2011 and 2012, combined model, Table 16).

Based on electrofishing, Chinook salmon juvenile abundance is higher at 10:1 Slope/Bench Sites than both Naturalized Sites and other Mitigated Repair Site Types. The strong association between Chinook salmon juveniles and 10:1 Slope/Bench Site Types has been consistent with results based on identical survey design from the previous year (FISHBIO 2012), though slightly different from habitat preferences reported by HTH & PRBO (2011), who found that though abundance of juvenile Chinook salmon was generally highest at 10:1 Slope/Bench sites early in the season (March), higher densities were observed at Wetland Trench Sites during later surveys (April). Unlike previous years (2009 and 2010, a dry and below normal water year, respectively) when higher abundance of Chinook salmon fry and juveniles was associated with Wetland Trench Site Types compared to other site types (H.T. Harvey & Associates and PRBO 2011), lower abundance of juveniles was associated with this site type in 2011 and 2012 (a wet and below normal water year, respectively).

Predation risk to Chinook salmon fry and juveniles is highest at Wetland Trench Sites, and predator abundance was higher at all Mitigated Repair Sites than at Naturalized Sites. The finding that combined predator abundance is significantly higher at Mitigated Repair Sites than at Naturalized Sites, and that the highest predator density is generally associated with Wetland Trench sites, followed by No Bench and 10:1 Slope/Bench locations, have remained consistent for combined survey data (2011 and 2012).

The overall effects of Mitigated Repair Site designs appear to outweigh the importance of individual microhabitat features in determining predator abundance.

Similar to results from 2011, none of the habitat variables analyzed in this study appeared to be consistently associated with predator abundance. Though steep banks (positive effects of bank slope) and with uniform current (negative association with current gradient) are considered significant predictors of predator abundance in the long-term survey dataset when repair type is not considered in the model, these effects become non-significant upon inclusion of repair design in GLM analysis.

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Appendix A. Environmental conditions at each sampling site during March 18-20, April 2-4, and April 16-18 2012 surveys.

Site ID	Date	Turbidity (ntu)	Temp (°C)	DO (mg/L)	Conductivity (µS/cm)	Salinity (ppmS)	Secchi (ft.)
24.0L	20-Mar	46.8	11.2	9.95	114.4	52.4	0.75
	4-Apr	44.8	12.8	10.2	128.8	64.9	1
	18-Apr	40.7	14.8	9.84	138.3	69	1.3
24.0R	20-Mar	46.8	11.2	9.95	114.4	52.4	0.75
	4-Apr	44.8	12.8	10.2	128.8	64.9	1
	18-Apr	40.7	14.8	9.84	138.3	69	1.3
25.0L	20-Mar	46.1	11.2	10	121.9	60.8	0.75
	4-Apr	44.9	12.8	10.21	120.9	59.8	0.75
	18-Apr	33.9	14.6	9.88	136.1	67.8	1.5
26.9L	20-Mar	36.3	11.1	10	152.9	64.5	0.75
	4-Apr	40.3	12.7	10.35	119.3	59.8	0.75
	18-Apr	33.9	14.5	9.92	142.1	74.2	1.5
34.5R	19-Mar	48.3	11.7	9.9	104.4	51.7	0.6
	3-Apr	50.8	13.1	10.26	123	60.3	0.75
	17-Apr	30.3	14.9	9.99	138.5	69.6	1
38.7R	19-Mar	44.8	11.4	10.04	114.9	56	0.75
	3-Apr	52.4	13.3	10.15	123.2	62.9	0.75
	17-Apr	33.4	15.0	10.01	135.8	66.6	1.25
43.7R	19-Mar	43.3	11.3	10.09	118.8	53.2	0.75
	3-Apr	56.3	13.7	10.15	121.2	62.5	0.75
	17-Apr	35.2	14.9	10.08	127.1	64.4	1.4
44.7R	19-Mar	41.2	11.2	10.03	115.6	56.3	0.75
	3-Apr	58.5	13.2	10.2	135.1	67.7	0.75
	17-Apr	29.2	14.7	10.12	132.1	65.8	1.6
47.0L	19-Mar	42.4	10.6	10.18	109.4	53	1
	3-Apr	58.1	13.3	10.24	129.4	64.7	1
	17-Apr	28.2	14.6	10.17	139	69.6	1.1
47.9R	19-Mar	48.1	10.8	10.26	107.4	55	0.75
	3-Apr	62.6	13.2	10.2	124.5	63.2	1
	17-Apr	29.9	14.8	10.09	121.9	56.7	1.4
50.2L	19-Mar	46.7	11.6	10.06	130.8	64.3	0.8
	3-Apr	47.8	12.8	10.37	126.3	59.2	1
	17-Apr	33.2	14.3	15.21	117.2	60.1	1.3
50.3R	18-Mar	54.5	11.2	10.24	110.7	54	0.5
	3-Apr	47.8	12.8	10.37	126.3	59.2	1
	17-Apr	33.2	14.3	10.21	117.2	60.1	1.3
56.0R	18-Mar	51.9	11.3	10.21	104.9	49.2	0.5
	3-Apr	64.9	12.4	10.5	145.5	69.9	1
	17-Apr	32.1	14.2	10.26	121.1	62.7	1

Site ID	Date	Turbidity (ntu)	Temp (°C)	DO (mg/L)	Conductivity (µS/cm)	Salinity (ppmS)	Secchi (feet)
69.1L	18-Mar	56.9	11.2	10.21	103.7	50.1	0.75
	2-Apr	51.3	13.3	10.23	143.8	72.9	1
	16-Apr	38.2	14.4	10.18	133.4	65.6	1.1
69.9R	18-Mar	48.3	11.7	10.19	99.6	46.8	0.5
	2-Apr	39.1	13.3	10.13	164.2	69.5	1
	16-Apr	41.7	14.3	10.24	139.9	72.8	1.1
72.2R	18-Mar	50.7	11.1	10.37	104.5	48.3	0.75
	2-Apr	44.2	14.9	10.2	138.9	67.8	1
	16-Apr	45.2	14.6	10.19	140	72.5	1
85.6L	17-Mar	36.7	11.7	10.16	132.1	65.9	1
	2-Apr	70.4	12.4	10.4	133.5	66	0.5
	16-Apr	50.6	13.3	10.56	124.4	60.3	1
85.6R	17-Mar	36.7	11.7	10.16	132.1	65.9	1
	2-Apr	70.4	12.4	10.4	133.5	66	0.5
	16-Apr	50.6	13.3	10.56	124.4	60.3	1
87.0L	17-Mar	47.9	11.7	10.16	118.9	59.7	1
	2-Apr	60.9	12.3	10.45	132.5	67.5	0.5
	16-Apr	48.1	14.2	10.5	127.3	62.5	1
90.8L	17-Mar	41.7	11.6	10.19	143.3	68.8	0.75
	2-Apr	74.7	12.2	10.48	134.5	69	0.5
	16-Apr	49.9	12.9	10.6	131.5	63.9	1.1
90.8R	17-Mar	41.7	11.6	10.19	143.3	68.8	0.75
	2-Apr	74.7	12.2	10.48	134.5	69	0.5
	16-Apr	49.9	12.9	10.6	131.5	63.9	1.1
91.9R	17-Mar	48.8	11.5	10.22	126.3	69.5	0.75
	2-Apr	64.7	11.6	10.79	168.9	68.8	0.5
	16-Apr	46.9	12.9	10.62	119.7	62.4	1

Appendix B. Number of species captured at each sampling site during March 17-19, April 2-4 and April 16-18, 2012 surveys.

Table B-1. Number of focus and predator species captured at each sampling site during March 17-19, April 2-4 and April 16-18, 2012 surveys.

Site ID	Date	Focus species			Predator species			
		Chinook juvenile	<i>O. mykiss</i>	Delta smelt	Sacramento pikeminnow	Striped bass	Black bass	Inland silverside
24.0L	20 Mar	33	0	0	1	1	2	15
	4 Apr	73	0	1	2	0	0	6
	18 Apr	29	0	0	6	0	3	4
24.0R	20 Mar	15	0	0	1	0	5	0
	4 Apr	7	0	0	0	0	9	0
	18 Apr	0	0	0	0	0	5	1
25.0L	20 Mar	28	0	5	0	0	0	8
	4 Apr	62	0	6	0	0	3	7
	18 Apr	21	0	0	4	0	6	1
26.9L	20 Mar	13	1	0	1	0	7	13
	4 Apr	8	0	1	1	0	12	3
	18 Apr	4	0	0	0	0	16	0
34.5R	19 Mar	34	0	2	0	1	4	3
	3 Apr	26	0	0	4	0	4	1
	17 Apr	9	0	2	1	0	2	0
38.7R	19 Mar	12	0	0	1	0	4	0
	3 Apr	11	0	0	0	0	3	0
	17 Apr	2	0	2	0	0	3	0
43.7R	19 Mar	13	1	2	0	0	0	2
	3 Apr	38	0	6	2	0	1	0
	17 Apr	36	0	8	5	0	1	0
44.7R	19 Mar	8	0	3	0	0	2	3
	3 Apr	6	0	0	7	1	2	3
	17 Apr	8	0	0	2	1	5	0

Site ID	Date	Focus species			Predator species			
		Chinook juvenile	<i>O. mykiss</i>	Delta smelt	Sacramento pikeminnow	Striped bass	Black bass	Inland silverside
47.0L	19 Mar	1	0	1	0	0	0	4
	3 Apr	8	0	3	6	0	2	10
	17 Apr	9	0	0	1	0	8	1
47.9R	19 Mar	8	0	3	1	0	2	0
	3 Apr	24	0	2	3	1	4	0
	17 Apr	9	0	0	0	0	5	0
50.2L	19 Mar	1	1	0	2	0	2	2
	3 Apr	11	0	0	4	0	4	0
	17 Apr	10	0	0	0	1	4	2
50.3R	18 Mar	13	1	1	2	0	0	1
	3 Apr	11	0	1	0	0	0	1
	17 Apr	6	0	0	0	0	4	1
56.0R	18 Mar	10	0	0	2	0	2	0
	3 Apr	9	0	0	6	0	4	0
	17 Apr	8	0	0	1	0	5	0
69.1L	18 Mar	20	2	0	1	0	0	11
	2 Apr	10	0	0	2	0	2	1
	16 Apr	14	0	0	0	0	4	0
69.9R	18 Mar	16	5	0	11	0	11	3
	3 Apr	10	0	0	0	0	0	1
	17 Apr	7	1	0	0	0	1	0
72.2R	18 Mar	11	0	0	1	0	0	5
	2 Apr	5	2	0	9	0	4	0
	16 Apr	7	0	0	15	0	1	0
85.6L	17 Mar	8	1	0	1	0	0	4
	2 Apr	12	0	0	0	0	0	0
	16 Apr	10	0	0	1	0	1	1
85.6R	17 Mar	7	1	0	1	1	2	0
	2 Apr	0	0	0	0	0	0	0
	16 Apr	6	0	0	1	0	1	0

Site ID	Date	Focus species			Predator species			
		Chinook juvenile	<i>O. mykiss</i>	Delta smelt	Sacramento pikeminnow	Striped bass	Black bass	Inland silverside
87.0L	17 Mar	24	0	0	2	0	3	0
	2 Apr	11	0	0	10	0	1	0
	16 Apr	2	0	0	3	0	1	0
90.8L	17 Mar	3	1	0	1	0	0	0
	2 Apr	9	0	0	2	0	0	0
	16 Apr	4	0	0	4	0	0	0
90.8R	17 Mar	5	0	0	1	0	0	2
	2 Apr	0	1	0	0	0	0	0
	16 Apr	1	1	0	0	0	0	0
91.9R	17 Mar	6	0	0	3	0	0	0
	2 Apr	5	0	0	3	0	0	0
	16 Apr	2	0	0	2	2	0	0

Table B-2. Number of other species captured at each sampling site during March 17-19, April 2-4 and April 16-18, 2012 surveys.

Site	American Shad	Bluegill sunfish	Black bullhead	Black crappie	Carp	Fathead minnow	Goldfish	Green sunfish	Golden shiner	Hitch	Hardhead	Mosquito fish	Prickly sculpin	Redear sunfish	Red Shiner	Sac. Sucker	Sac. blackfish	Spittail	Threadfin shad	Tule perch	Warmouth	Wakasagi
March 17-19, 2012																						
24.0L	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	2	0	0
24.0R	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
25.0L	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	1	0	0
26.9L	0	3	0	0	0	0	0	0	2	0	0	0	2	0	0	0	0	1	0	8	0	0
34.5R	0	2	0	1	0	0	0	0	0	1	0	0	1	1	0	0	0	11	1	3	0	0
38.7R	0	0	0	1	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
43.7R	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0
44.7R	0	1	0	0	0	0	1	0	0	1	0	0	0	0	0	0	0	1	1	0	0	0
47.0L	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
47.9R	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	3	0	0
50.2L	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
50.3R	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	1	0	0
56.0R	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
69.1L	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	7	0	1	0	2
69.9R	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	1	0	11	0	2	0	1
72.2R	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	0	2	1	0	0	0
85.6L	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0
85.6R	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
87.0L	0	0	0	0	4	0	0	0	0	0	0	0	0	0	0	6	0	1	0	4	0	0
90.8L	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
90.8R	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0	0
91.9R	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Total	0	8	0	2	5	0	1	0	2	5	0	1	3	1	0	12	0	41	3	34	0	3

Table B-2. cont.

Site	American Shad	Bluegill sunfish	Black bullhead	Black crappie	Carp	Fathead minnow	Goldfish	Green sunfish	Golden shiner	Hitch	Hardhead	Mosquito fish	Prickly sculpin	Redear sunfish	Red Shiner	Sac. Sucker	Sac. blackfish	Spittail	Threadfin shad	Tule perch	Warmouth	Wakasagi
April 2 -4, 2012																						
24.0L	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	0	0	0	8	0	0
24.0R	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
25.0L	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0
26.9L	0	4	0	0	0	0	0	0	4	0	0	0	0	0	0	0	0	0	0	2	0	2
34.5R	0	1	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0	1	0	0
38.7R	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
43.7R	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	6	0	0	0	0
44.7R	0	3	0	0	0	0	0	0	1	0	0	0	1	1	0	0	0	0	0	3	0	0
47.0L	0	6	1	0	0	0	0	0	1	0	0	0	2	0	0	0	0	1	0	3	1	0
47.9R	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
50.2L	0	2	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	3	0	0
50.3R	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	3	0	0	0	1
56.0R	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0	0
69.1L	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0
69.9R	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0	0	0	0
72.2R	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0	7	0	0
85.6L	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	1	2	2	0	0
85.6R	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
87.0L	0	1	0	0	3	0	1	1	1	0	0	0	0	0	1	0	0	0	0	11	0	0
90.8L	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
90.8R	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	0	0
91.9R	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Total	0	21	1	1	5	1	1	3	9	0	0	0	5	1	1	2	0	21	2	50	1	3

Table B-2. cont.

Site	American Shad	Bluegill sunfish	Black bullhead	Black crappie	Carp	Fathead minnow	Goldfish	Green sunfish	Golden shiner	Hitch	Hardhead	Mosquito fish	Prickly sculpin	Redear sunfish	Red Shiner	Sac. Sucker	Sac. blackfish	Splittail	Threadfin shad	Tule perch	Warmouth	Wakasagi
April 16 -18, 2012																						
24.0L	0	0	0	0	0	0	0	0	0	2	0	0	1	0	0	0	0	0	0	2	0	0
24.0R	0	0	0	0	0	0	0	1	0	0	0	0	2	0	0	1	0	0	0	0	0	0
25.0L	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	1	0	0
26.9L	0	2	0	0	0	0	0	1	0	0	0	0	2	0	0	0	0	0	0	1	0	0
34.5R	0	1	0	0	0	0	0	0	0	4	0	0	1	0	0	2	0	0	0	0	0	0
38.7R	0	0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	0	0
43.7R	2	0	0	0	0	0	0	0	0	0	0	0	1	0	0	6	0	5	0	1	0	0
44.7R	0	0	0	0	0	0	0	0	0	0	0	0	6	0	0	0	0	0	0	1	0	0
47.0L	0	2	0	0	0	0	0	2	0	1	0	0	5	2	0	3	0	0	0	1	0	0
47.9R	0	2	0	0	1	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0
50.2L	0	0	0	0	3	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	0	0
50.3R	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
56.0R	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	1	0	0	0	0	0	0
69.1L	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0
69.9R	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0
72.2R	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	2	0	4	0	0
85.6L	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0	0
85.6R	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
87.0L	0	0	0	0	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	6	0	0
90.8L	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	0	0
90.8R	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	1	0	0	0	4	0	0
91.9R	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	2	2	0	0
Total	2	8	0	0	14	0	0	4	0	8	1	0	27	2	0	15	1	10	3	30	0	0